Finite-Step Sinkhorn Certificates for Nonnegative Stream-Mixing Products

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Abstract

We develop deterministic finite-step bounds for products of nonnegative stream-mixing matrices in expanded-stream residual architectures, with Manifold-Constrained Hyper-Connections as the motivating example. We study the depth-wise mixer product

$$T_L = H_{\text{res}}^{(L)} H_{\text{res}}^{(L-1)} \cdots H_{\text{res}}^{(1)}$$

and bound its forward and backward induced-norm gains. For entrywise nonnegative mixers, row and column marginal defects directly control the corresponding induced norms. These certificates concern the mixer pathway; full-network stability additionally depends on the nonlinear residual branches. In the column-then-row Sinkhorn order, exact row sums give layerwise forward control, while the output column defect gives

$$\|T_L\|_1 \le \min\Big\{\prod_{\ell=1}^{L}(1+\varepsilon_c^{(\ell)}),\, n\Big\},$$

where n is a structural ceiling. We complement this product bound with a telescoping refinement and a depth-uniform Dobrushin certificate under uniform scrambling. A Hilbert-projective argument converts logit dynamic range, temperature, and a finite Sinkhorn step cap into a conservative backward mixer-path budget, while matrix-specific local analysis gives the asymptotic rate $\sigma_2(P)^2$. Numerical diagnostics examine marginal defects, realized and certified gains, finite-step calibration, cap failures, and normalization cost in matched and stress-test settings.

1 Introduction

A conventional residual block propagates one feature state through an identity skip connection and a nonlinear update (He et al., 2016). An expanded-stream residual architecture instead carries n parallel feature states. Its nonlinear branch reads from and writes to aggregates of those states, while a trainable matrix $H_{\text{res}}^{(\ell)} \in \mathbb{R}^{n\times n}$ routes the skip pathway between streams at layer ℓ. The identity matrix recovers an unmixed skip pathway; a nontrivial mixer allows information to be redistributed before the next block. This is the role played by the stream mixers in hyper-connections and mHC (Xie et al., 2025).

The mixer is algebraically similar to an attention matrix, but it acts on a small architectural stream index rather than on token positions: self-attention mixes data-dependent token representations (Vaswani et al., 2017), whereas the object studied here routes residual states across depth. Repeated routing produces the depth-wise mixing pathway

$$T_L := H_{\text{res}}^{(L)} H_{\text{res}}^{(L-1)} \cdots H_{\text{res}}^{(1)}. \tag{1.1}$$

The author used ChatGPT and received coding assistance from OpenAI Codex. The author takes responsibility for all content.

Large induced norms of T_L can amplify activations or backpropagated signals through this pathway, even when feature-scale normalization is present. This separates a stream-routing question from the behavior of the nonlinear feature branch and is the reason we study the product directly.

Nonnegative row-stochastic mixers are natural here, just as attention weights are commonly normalized by rows, but row-stochasticity by itself is insufficient for backward control. For example, $H = \mathbf{1}\mathbf{e}_1^\top$ has $\|H\|_\infty = 1$ but $\|H\|_1 = n$. Thus a row-normalized mixer can preserve forward ℓ_∞ gain while amplifying backward signals at the stream-expansion scale.

Sinkhorn–Knopp normalization alternately rescales the columns and rows of a positive matrix toward a doubly stochastic matrix. Practical neural modules use a finite number of these inexpensive scaling steps. In the column-then-row schedule studied here, the final row sums are exact in ideal arithmetic, while the remaining column defect determines the backward induced-norm error. A fixed iteration count alone, however, does not state what depth-wise budget has been certified.

The motivating architecture class is Manifold-Constrained Hyper-Connections (mHC), which parameterizes nonnegative mixers and pushes them toward the Birkhoff polytope of doubly stochastic matrices. For the iterative parameterization, the implementation-relevant object is a positive logit matrix followed by finitely many Sinkhorn–Knopp normalizations (Sinkhorn and Knopp, 1967; Knight, 2008). Given a depth L, a backward mixer-path budget B_1, and a step cap $t_{\max}$, we ask what finite-step marginal tolerance suffices to certify $\|T_L\|_1 \le B_1$.

The analysis proceeds from quantities available in the implementation. Measured column defects control the induced ℓ_1 norm, Hilbert-metric contraction relates those defects to logit dynamic range and the number of Sinkhorn steps, and a reverse vector propagation computes the exact realized backward gain. In the ideal column-then-row order, every output has exact row sums, so forward $A_{\max}$ mixer-path gain is controlled layerwise; the remaining issue is backward ℓ_1 gain.

Sinkhorn–Knopp scaling and matrix balancing are classical (Sinkhorn and Knopp, 1967; Franklin and Lorenz, 1989; Knight, 2008), and Hilbert's projective metric gives a standard contraction framework for positive maps (Birkhoff, 1957; Bushell, 1973). In machine learning, Sinkhorn iterations are used in entropic optimal transport and Sinkhorn distances (Cuturi, 2013; Peyré and Cuturi, 2019), and Sinkhorn-style normalization has been studied for attention (Sander et al., 2022; Shahbazi et al., 2025). Here Sinkhorn normalization is used to certify finite-step error budgets for products of nonnegative stream mixers. Norm control for residual dynamics is also related to stable deep architectures and continuous-depth models (Haber and Ruthotto, 2018; Chen et al., 2018); our certificates target the finite-step normalization error of the nonnegative mixing pathway.

An alternative is to avoid finite normalization error by parameterizing an exactly doubly stochastic mixer. In particular, mHC-lite constructs mixers as convex combinations of permutation matrices (Yang and Gao, 2026). That approach and the present one address the same approximation gap in different ways: an exact parameterization removes it by construction, whereas our results retain iterative Sinkhorn normalization and certify its realized finite-step error and compute budget.

The ingredients—Sinkhorn scaling, Hilbert-projective contraction, and induced-norm identities for nonnegative matrices—are classical. The contribution here is their schedule-aware combination for finite products of neural stream mixers. The analysis tracks the output half-step in the column-then-row schedule and propagates the measured column defect through product, structural, telescoping, and Dobrushin bounds, leading to calibration and stopping rules stated directly in terms of measurable finite-step errors.

Scope. The results control the mixer product T_L in induced ℓ_∞ and ℓ_1 matrix norms, equivalently the forward and backward streamwise $A_{\max}$ gains of the mixing pathway. Full-block bounds additionally require control of the nonlinear branch Jacobians; Appendix C records the corresponding composition estimates. Throughout the paper, "certificate" refers to this mixer-path control.

Contributions.

- **Mixer-product certificates.** For entrywise nonnegative mixers, row and column marginal errors directly control $\|H\|_\infty$ and $\|H\|_1$, giving depth-wise forward and backward $A_{\max}$ budgets for T_L (Proposition 3.1). We also record the exact realized backward-gain diagnostic by vector propagation (Equation (3.4)).
- **Sharper product controls.** The backward gain of any product of nonnegative row-stochastic mixers is capped by the expansion rate, $\|T_L\|_1 \le n$ (Proposition 3.2), independently of the Sinkhorn schedule. A telescoping prefix-gain bound refines the raw product certificate (Lemma 3.3), and a layerwise Dobrushin bound gives a depth-uniform certificate under uniform scrambling (Proposition 3.4).
- **Finite-step Sinkhorn calibration.** Applying Hilbert-metric contraction to the column-then-row normalization schedule, we derive a column-marginal estimate with the proper one-half-step loss at the output. A heterogeneous master certificate gathers the product, structural, telescoping, and Dobrushin alternatives for layer-dependent matrices and step counts (Corollary 4.4). The common-κ, common-step rule then maps $(\kappa, t_{\max})$ to a certified tolerance and a depth-wise backward budget on $\|T_L\|_1$ (Theorem 4.3 and Theorem 4.7).
- **Diagnostics and local rates.** For a fixed positive matrix, the local full-step Sinkhorn rate is $\sigma_2(P)^2$, used as a predictor rather than a global certificate. The experiments track marginal defects, product certificates, exact propagated gains, cap failures, adaptive stopping, and controller costs.

2 Preliminaries and notation

Notation summary

Symbol	Meaning
n	number of streams / expansion rate
C	feature width per stream
$H_{\text{res}}^{(\ell)}$	stream-mixing matrix at layer ℓ
T_L	depth-wise mixer product $H_{\text{res}}^{(L)} \cdots H_{\text{res}}^{(1)}$
$A = S^{(0)}$	positive pre-normalized matrix before Sinkhorn scaling
T_c, T_r	column and row normalization maps
$S^{(s)}$	matrix after s alternating normalization half-steps
t	number of full column-then-row Sinkhorn steps; output is $S^{(2t)}$
$\varepsilon_r, \varepsilon_c$	row and column marginal errors
κ	entrywise dynamic range $\max_{ij} A_{ij} / \min_{ij} A_{ij}$
$\Delta(A)$	projective diameter of A
$\tau(A)$	Birkhoff contraction factor $\tanh(\Delta(A)/4)$
q	dynamic-range bound $(\kappa-1)/(\kappa+1)$ for $\tau(A)$
P	limiting doubly stochastic Sinkhorn scaling of A
ρ_*	local full-step rate $\sigma_2(P)^2$
$A_{\max}$	streamwise coordinate maximum functional

2.1 Stream convention and $A_{\max}$ gains

Fix an expansion rate $n \ge 1$ and feature width $C \ge 1$. At depth ℓ, the state is a matrix $x_\ell \in \mathbb{R}^{n \times C}$ whose rows index streams and whose columns index features. A hyper-connected residual layer has the form

$$x_\ell = H_{\text{res}}^{(\ell)} x_{\ell-1} + (H_{\text{post}}^{(\ell)})^\top F\big(H_{\text{pre}}^{(\ell)} x_{\ell-1}, W_\ell\big), \qquad \ell = 1, \dots, L, \tag{2.1}$$

where $H_{\text{res}}^{(\ell)} \in \mathbb{R}^{n \times n}$ mixes streams, $H_{\text{pre}}^{(\ell)}, H_{\text{post}}^{(\ell)} \in \mathbb{R}^{1 \times n}$ are pre/post aggregation row vectors, and $F(\cdot, W_\ell) : \mathbb{R}^{1 \times C} \to \mathbb{R}^{1 \times C}$ is a nonlinear feature block. The certificate below concerns the stream-mixer product generated by $H_{\text{res}}^{(\ell)}$.

For $M \in \mathbb{R}^{n \times n}$ we use the matrix norms induced by the vector ℓ_∞ and ℓ_1 norms:

$$\|M\|_\infty := \max_i \sum_j |M_{ij}|, \qquad \|M\|_1 := \max_j \sum_i |M_{ij}|. \tag{2.2}$$

Throughout, $\|\cdot\|_1$ and $\|\cdot\|_\infty$ refer to these matrix norms; for vector norms we write $\|\cdot\|_{1,\mathrm{vec}}$ and $\|\cdot\|_{\infty,\mathrm{vec}}$.

Definition 2.1 (Streamwise $A_{\max}$ functional)**.** For $X \in \mathbb{R}^{n\times C}$, define

$$A_{\max}(X) := \max_{1\le i\le n} \|X_{i,:}\|_{\infty,\mathrm{vec}} = \max_{i,c} |X_{ic}|. \tag{2.3}$$

For $H \in \mathbb{R}^{n\times n}$ and $X, G \in \mathbb{R}^{n\times C}$,

$$A_{\max}(HX) \le \|H\|_\infty \, A_{\max}(X), \tag{2.4}$$

while backpropagation through the stream-mixing map uses the adjoint $G \mapsto H^\top G$ and satisfies

$$A_{\max}(H^\top G) \le \|H\|_1 \, A_{\max}(G). \tag{2.5}$$

Thus forward streamwise $A_{\max}$ gain is governed by $\|H\|_\infty$, while the corresponding backward gain is governed by $\|H\|_1$. If $H \ge 0$, these norms are just the maximum row sum and maximum column sum.

Approximate doubly stochasticity. The Birkhoff polytope of doubly stochastic matrices is

$$\mathcal{B}_n := \left\{ P \in \mathbb{R}^{n\times n} : \; P_{ij} \ge 0, \; \sum_{j=1}^{n} P_{ij} = 1 \; \forall i, \; \sum_{i=1}^{n} P_{ij} = 1 \; \forall j \right\}. \tag{2.6}$$

Since the mixers use a finite number of Sinkhorn–Knopp steps, we quantify closeness to $\mathcal{B}_n$ by marginal errors.

Definition 2.2 (Approximate doubly stochasticity)**.** Let $H \in \mathbb{R}^{n\times n}$ be entrywise nonnegative. We say that H is $(\varepsilon_r, \varepsilon_c)$-doubly stochastic if

$$\max_i \left| \sum_j H_{ij} - 1 \right| \le \varepsilon_r, \qquad \max_j \left| \sum_i H_{ij} - 1 \right| \le \varepsilon_c. \tag{2.7}$$

3 Depth-wise $A_{\max}$ certificates

This section converts per-layer marginal errors into worst-case bounds on forward and backward $A_{\max}$ gains for the depth-wise mixer product. For entrywise nonnegative mixers, marginal control directly bounds the induced ℓ_∞ and ℓ_1 matrix norms, with no additional structural assumptions.

If $H \ge 0$ is $(\varepsilon_r, \varepsilon_c)$-doubly stochastic in the sense of Definition 2.2, then

$$\|H\|_\infty \le 1 + \varepsilon_r, \qquad \|H\|_1 \le 1 + \varepsilon_c, \tag{3.1}$$

since for nonnegative H the norms reduce to maximum row sum and maximum column sum.

Proposition 3.1 (Marginal-error depth certificate)**.** *Let $H_{\mathrm{res}}^{(1)}, \dots, H_{\mathrm{res}}^{(L)} \in \mathbb{R}^{n\times n}$ be entrywise nonnegative and define T_L as in Equation (1.1). Assume each $H_{\mathrm{res}}^{(\ell)}$ is $(\varepsilon_r^{(\ell)}, \varepsilon_c^{(\ell)})$-doubly stochastic. Then*

$$\|T_L\|_\infty \le \prod_{\ell=1}^{L} (1 + \varepsilon_r^{(\ell)}), \qquad \|T_L\|_1 \le \prod_{\ell=1}^{L} (1 + \varepsilon_c^{(\ell)}). \tag{3.2}$$

In particular, marginal error certificates propagate multiplicatively across depth into a depth-wise $A_{\max}$ budget for the mixer pathway.

Proof. By submultiplicativity of induced norms and (3.1), $\|T_L\|_\infty \le \prod_{\ell=1}^{L} \left\| H_{\mathrm{res}}^{(\ell)} \right\|_\infty \le \prod_{\ell=1}^{L} (1 + \varepsilon_r^{(\ell)})$, and similarly for $\|\cdot\|_1$. □

Exact realized backward gain. For realized nonnegative mixers $H^{(1)}, \dots, H^{(L)}$, set $T_L = H^{(L)} \cdots H^{(1)}$. The exact backward mixing gain is obtained without forming T_L:

$$w_L = \mathbf{1}, \qquad w_{\ell-1} = (H^{(\ell)})^\top w_\ell, \qquad \ell = L, L-1, \dots, 1. \tag{3.3}$$

Then

$$w_0 = T_L^\top \mathbf{1}, \qquad \|T_L\|_1 = \|w_0\|_{\infty,\mathrm{vec}}. \tag{3.4}$$

Indeed, for a nonnegative matrix the induced ℓ_1 norm is the maximum column sum. This exact diagnostic can be substantially smaller than the layerwise product bound.

3.1 A structural ceiling and a telescoping refinement

The product certificate of Proposition 3.1 can grow with depth. For row-stochastic nonnegative mixers, however, the true backward gain is capped by the expansion rate. The structural ceiling follows from row-stochasticity. A telescoping argument then interpolates between the layerwise product certificate and the exact propagated gain.

Proposition 3.2 (Structural backward ceiling at the expansion rate)**.** *Let $H^{(1)}, \dots, H^{(L)} \in \mathbb{R}^{n\times n}$ be entrywise nonnegative and set $T_L = H^{(L)} \cdots H^{(1)}$.*

(i) If every factor has exact row sums, $H^{(\ell)}\mathbf{1} = \mathbf{1}$ (as in the exact column–then–row output (4.8)), then T_L is nonnegative and row-stochastic, and

$$1 \le \|T_L\|_1 \le n. \tag{3.5}$$

(ii) More generally, if each $H^{(\ell)}$ is $(\varepsilon_r^{(\ell)}, \varepsilon_c^{(\ell)})$-doubly stochastic, then

$$\|T_L\|_1 \le n \prod_{\ell=1}^{L} \big(1 + \varepsilon_r^{(\ell)}\big). \tag{3.6}$$

Combining (3.6) with Proposition 3.1,

$$\|T_L\|_1 \le \min\left\{ \prod_{\ell=1}^{L} \big(1 + \varepsilon_c^{(\ell)}\big),\ n \prod_{\ell=1}^{L} \big(1 + \varepsilon_r^{(\ell)}\big) \right\}, \tag{3.7}$$

and in the ideal column–then–row order $\|T_L\|_1 \le \min\big\{\prod_\ell (1+\varepsilon_c^{(\ell)}),\, n\big\}$.

Proof. A product of nonnegative row-stochastic matrices is nonnegative and row-stochastic, so $T_L\mathbf{1} = \mathbf{1}$ and the total mass is $\mathbf{1}^\top T_L \mathbf{1} = \mathbf{1}^\top \mathbf{1} = n$. For a nonnegative matrix $\|T_L\|_1$ equals the maximum column sum, which lies between the mean column sum $n/n = 1$ and the total mass n, giving (3.5). For (ii), nonnegativity together with $H^{(\ell)}\mathbf{1} \le (1+\varepsilon_r^{(\ell)})\mathbf{1}$ entrywise gives, by induction on the factors, $T_L\mathbf{1} \le \prod_\ell (1+\varepsilon_r^{(\ell)})\,\mathbf{1}$ entrywise; thus $\mathbf{1}^\top T_L \mathbf{1} \le n \prod_\ell (1+\varepsilon_r^{(\ell)})$, and the maximum column sum is at most the total mass. □

When the product certificate improves the structural ceiling. For nonnegative row-stochastic mixers, $\|T_L\|_1 \le n$ holds independently of the Sinkhorn schedule. Thus the finite-step product certificate is useful as a sharper worst-case bound when

$$\prod_{\ell=1}^{L} (1 + \varepsilon_c^{(\ell)}) < n.$$

If this product is at least n, the structural ceiling already gives the smaller bound.

Lemma 3.3 (Telescoping backward certificate)**.** *Assume the ideal column–then–row order, so each $H^{(\ell)}$ is nonnegative with exact row sums. Write the prefix products $T_0 := I$ and $T_k := H^{(k)} \cdots H^{(1)}$, and let $\varepsilon_c^{(\ell)} = \|\mathbf{1}^\top H^{(\ell)} - \mathbf{1}^\top\|_{\infty,\mathrm{vec}}$. Then*

$$\|T_L\|_1 \le 1 + \sum_{\ell=1}^{L} \|T_{\ell-1}\|_1 \, \varepsilon_c^{(\ell)}. \tag{3.8}$$

Bounding the realized prefix gains by Proposition 3.2, $\|T_{\ell-1}\|_1 \le \min\{\prod_{j<\ell}(1+\varepsilon_c^{(j)}), n\}$, gives the a priori refinement

$$\|T_L\|_1 \le 1 + \sum_{\ell=1}^{L} \min\Big\{\textstyle\prod_{j<\ell}(1+\varepsilon_c^{(j)}),\, n\Big\}\, \varepsilon_c^{(\ell)} \le 1 + n \textstyle\sum_{\ell=1}^{L} \varepsilon_c^{(\ell)}. \tag{3.9}$$

Substituting instead the multiplicative bound $\|T_{\ell-1}\|_1 \le \prod_{j<\ell}(1+\varepsilon_c^{(j)})$ recovers exactly the product certificate $\prod_\ell(1+\varepsilon_c^{(\ell)})$ of Proposition 3.1. The telescoping bound does not exceed the product certificate, and it is strictly smaller whenever the realized prefix gains grow sub-multiplicatively.

Proof. Let $w_L = \mathbf{1}$ and $w_{\ell-1} = (H^{(\ell)})^\top w_\ell$, so $w_0 = T_L^\top \mathbf{1}$ and $\|T_L\|_1 = \|w_0\|_{\infty,\mathrm{vec}}$ by Equation (3.4). Write $w_\ell = \mathbf{1} + d_\ell$. Since $(H^{(\ell)})^\top \mathbf{1} = \mathbf{1} + c_\ell$ with $c_\ell := (H^{(\ell)})^\top \mathbf{1} - \mathbf{1}$, $\|c_\ell\|_{\infty,\mathrm{vec}} = \varepsilon_c^{(\ell)}$, and $\mathbf{1}^\top c_\ell = \mathbf{1}^\top (H^{(\ell)})^\top \mathbf{1} - n = (H^{(\ell)}\mathbf{1})^\top \mathbf{1} - n = 0$ by exact row sums, the recursion gives

$$d_{\ell-1} = c_\ell + (H^{(\ell)})^\top d_\ell, \qquad d_L = 0,$$

with every $d_\ell \in \mathbf{1}^\perp$. Unrolling,

$$d_0 = \sum_{\ell=1}^{L} (H^{(1)})^\top \cdots (H^{(\ell-1)})^\top c_\ell = \sum_{\ell=1}^{L} T_{\ell-1}^\top c_\ell,$$

using $(H^{(1)})^\top \cdots (H^{(\ell-1)})^\top = (H^{(\ell-1)} \cdots H^{(1)})^\top = T_{\ell-1}^\top$. Because $w_0 = \mathbf{1} + d_0 \ge 0$ entrywise, $\|T_L\|_1 = \max_i (w_0)_i = 1 + \max_i (d_0)_i \le 1 + \|d_0\|_{\infty,\mathrm{vec}}$, and

$$\|T_{\ell-1}^\top c_\ell\|_{\infty,\mathrm{vec}} \le \|T_{\ell-1}^\top\|_\infty \, \|c_\ell\|_{\infty,\mathrm{vec}} = \|T_{\ell-1}\|_1 \, \varepsilon_c^{(\ell)},$$

which gives (3.8). The algebraic identity $\prod_\ell(1+\varepsilon_c^{(\ell)}) = 1 + \sum_\ell \varepsilon_c^{(\ell)} \prod_{j<\ell}(1+\varepsilon_c^{(j)})$ shows the product certificate is exactly the case in which each prefix gain is replaced by its multiplicative upper bound. □

Computing the telescoping refinement. The ordered prefix gains in (3.8) can be computed by maintaining the transposed prefix matrix:

$$Q_0 = I, \qquad Q_\ell = Q_{\ell-1}(H^{(\ell)})^\top = T_\ell^\top, \qquad \|T_\ell\|_1 = \|Q_\ell \mathbf{1}\|_{\infty,\mathrm{vec}}.$$

For dense mixers this costs $O(Ln^3)$, whereas the a priori bound (3.9) uses only the marginal defects. The exact final gain in (3.4) costs $O(Ln^2)$ by a single reverse vector propagation. Thus the telescoping estimate is most useful when prefix products are already being formed or audited.

3.2 Depth-uniform control under scrambling

The product and telescoping certificates are worst-case bounds driven by marginal defects. Dobrushin contraction gives a complementary finite-horizon bound that retains the layerwise mixing coefficients. When the row-stochastic mixers are uniformly scrambling, the same argument gives a certificate independent of the depth.

Proposition 3.4 (Finite-horizon and depth-uniform Dobrushin certificate)**.** *Let $H^{(1)}, \dots, H^{(L)} \in \mathbb{R}^{n\times n}$ be entrywise nonnegative with exact row sums $H^{(\ell)}\mathbf{1} = \mathbf{1}$. For each layer define the column-defect vector*

$$c_\ell := (H^{(\ell)})^\top \mathbf{1} - \mathbf{1}$$

and write

$$\delta_\ell := \delta_{\mathrm{D}}(H^{(\ell)}) := \frac{1}{2} \max_{i,k} \sum_j |H^{(\ell)}_{ij} - H^{(\ell)}_{kj}|.$$

Then the depth-wise backward mixer gain satisfies the finite-horizon bound

$$\|T_L\|_1 \le 1 + \frac{1}{2} \sum_{\ell=1}^{L} \left(\prod_{j=1}^{\ell-1} \delta_j \right) \|c_\ell\|_{1,\mathrm{vec}}, \tag{3.10}$$

where the empty product is one. If $\bar{\delta} := \max_\ell \delta_\ell < 1$*, then*

$$\|T_L\|_1 \le 1 + \frac{\max_\ell \|c_\ell\|_{1,\mathrm{vec}}}{2} \frac{1 - \bar{\delta}^L}{1 - \bar{\delta}} \le 1 + \frac{\max_\ell \|c_\ell\|_{1,\mathrm{vec}}}{2(1 - \bar{\delta})}. \tag{3.11}$$

The final expression is independent of the depth L*.*

Coordinatewise versus total-variation defects. The Dobrushin estimate uses the total-variation defect $\|c_\ell\|_{1,\mathrm{vec}}$, whereas the coordinatewise marginal defect used in Proposition 3.1 is $\varepsilon_c^{(\ell)} = \|c_\ell\|_{\infty,\mathrm{vec}}$. Since $\mathbf{1}^\top c_\ell = 0$,

$$\|c_\ell\|_{1,\mathrm{vec}} \le 2 \left\lfloor \frac{n}{2} \right\rfloor \varepsilon_c^{(\ell)} \le n \varepsilon_c^{(\ell)}.$$

Thus the depth-uniform Dobrushin certificate may carry an explicit dependence on the number of streams when it is expressed only through coordinatewise marginal errors. In particular, if $\varepsilon_{\max} := \max_\ell \varepsilon_c^{(\ell)}$ and $\bar{\delta} < 1$, then

$$\|T_L\|_1 \le 1 + \left\lfloor \frac{n}{2} \right\rfloor \varepsilon_{\max} \frac{1 - \bar{\delta}^L}{1 - \bar{\delta}}. \tag{3.12}$$

Proof. Let $w_L = \mathbf{1}$ and $w_{\ell-1} = (H^{(\ell)})^\top w_\ell$. Write $w_\ell = \mathbf{1} + d_\ell$. Since each $H^{(\ell)}$ is row-stochastic, $\mathbf{1}^\top c_\ell = 0$, and the recursion gives

$$d_{\ell-1} = c_\ell + (H^{(\ell)})^\top d_\ell, \qquad d_L = 0.$$

For any zero-sum vector z, the Dobrushin coefficient is the total-variation contraction coefficient:

$$\|(H^{(\ell)})^\top z\|_{1,\mathrm{vec}} \le \delta_{\mathrm{D}}(H^{(\ell)}) \|z\|_{1,\mathrm{vec}}.$$

Unrolling the recursion gives

$$d_0 = c_1 + (H^{(1)})^\top c_2 + \cdots + (H^{(1)})^\top \cdots (H^{(L-1)})^\top c_L.$$

Repeated Dobrushin contraction gives

$$\|d_0\|_{1,\mathrm{vec}} \le \sum_{\ell=1}^{L} \left(\prod_{j=1}^{\ell-1} \delta_j \right) \|c_\ell\|_{1,\mathrm{vec}}.$$

Moreover, $\mathbf{1}^\top d_0 = 0$, so its largest coordinate is at most its total positive mass, namely $\frac{1}{2}\|d_0\|_{1,\mathrm{vec}}$. Since $w_0 = T_L^\top \mathbf{1} = \mathbf{1} + d_0$,

$$\|T_L\|_1 = \|w_0\|_{\infty,\mathrm{vec}} \le 1 + \frac{1}{2} \|d_0\|_{1,\mathrm{vec}},$$

which proves (3.10). Bounding each δ_j by $\bar{\delta}$, each defect by its layerwise maximum, and summing the finite geometric series proves (3.11). □

Why nonnegativity is essential. Proposition 3.1 uses entrywise nonnegativity through the elementary identities

$$\|H\|_\infty = \max_i \sum_j H_{ij}, \qquad \|H\|_1 = \max_j \sum_i H_{ij}, \qquad H \geq 0.$$

Without nonnegativity, exact row and column sums do not control induced norms. Indeed, for $N > 1$,

$$A_N = \begin{pmatrix} N & 1-N \\ 1-N & N \end{pmatrix}$$

has all row and column sums equal to one, but

$$\|A_N\|_\infty = \|A_N\|_1 = 2N - 1.$$

Since A_N is symmetric, also $\|A_N\|_2 = 2N - 1$. Thus marginal constraints alone do not prevent arbitrary amplification when signs are allowed. Positivity, enforced here by exponentiating logits, is structural for the marginal-error certificate.

The preceding estimates are the mixer-path certificates used throughout the paper. Appendix C records the corresponding Jacobian bounds when these mixer-path estimates are embedded into a full residual block.

4 Finite-step Sinkhorn: bounds and calibration

Proposition 3.1 bounds depth-wise mixer gains in terms of per-layer marginal errors. This section relates those errors to the finite-step Sinkhorn iterations used in the mHC-motivated mixer parameterization. Operationally, a target backward budget is first converted into per-layer column-defect tolerances; Sinkhorn iterations are then stopped when those measured tolerances are met, subject to a finite step cap. The resulting product certificate, structural ceiling, and exact propagated gain are reported separately.

Standing assumptions. Throughout Section 4, the pre-scaling matrix is strictly positive, $S^{(0)} \in \mathbb{R}^{n\times n}_{>0}$, and has bounded dynamic range

$$\kappa := \frac{\max_{i,j} S^{(0)}_{ij}}{\min_{i,j} S^{(0)}_{ij}} < \infty. \tag{4.1}$$

In applications this is enforced by clipped logits, for example $S^{(0)}_{ij} = \exp(\beta C_{ij})$ with $C_{ij} \in [-R, R]$. A full Sinkhorn step means column normalization followed by row normalization. If a layer uses $t^{(\ell)} \in \{1, \ldots, t_{\max}\}$ full steps, then

$$H^{(\ell)}_{\text{res}} = S^{(2t^{(\ell)})}. \tag{4.2}$$

The fixed-step variant uses $t^{(\ell)} = t_{\max}$; the early-stopped variant uses $t^{(\ell)} = t^{(\ell)}_{\text{stop}} \leq t_{\max}$. The theoretical statements in this section use exact row and column normalization; implemented variants are discussed below.

4.1 Alternating normalization and marginal errors

Given logits $\widetilde{H_{\text{res}}}^{(\ell)} \in \mathbb{R}^{n\times n}$, the mixer parameterization forms a positive matrix

$$S^{(0)} := \exp(\beta\, \widetilde{H_{\text{res}}}^{(\ell)}) \qquad \text{(entrywise)}, \tag{4.3}$$

where $\beta > 0$ is an optional temperature. We then apply alternating column and row normalizations:

$$S^{(s+1)} := \begin{cases} T_c(S^{(s)}) & s \text{ even}, \\ T_r(S^{(s)}) & s \text{ odd}, \end{cases} \qquad s = 0, 1, 2, \ldots \tag{4.4}$$

where T_c divides each column by its column sum and T_r divides each row by its row sum. After a column normalization we have exact column sums, and after a row normalization we have exact row sums:

$$\mathbf{1}^\top S^{(s)} = \mathbf{1}^\top \ \ (s \text{ odd}), \qquad S^{(s)}\mathbf{1} = \mathbf{1} \ \ (s \text{ even},\ s \geq 2). \tag{4.5}$$

Accordingly, the nontrivial marginal errors occur on alternating half-steps. We define

$$\varepsilon_r^{(s)} := \left\| S^{(s)}\mathbf{1} - \mathbf{1} \right\|_{\infty,\mathrm{vec}}, \qquad \varepsilon_c^{(s)} := \left\| \mathbf{1}^\top S^{(s)} - \mathbf{1}^\top \right\|_{\infty,\mathrm{vec}}. \tag{4.6}$$

If we run t *full* Sinkhorn steps in the column–then–row order, this corresponds to $2t$ alternating normalizations and the output

$$H_{\mathrm{res}}^{(\ell)} := S^{(2t)}. \tag{4.7}$$

We use either (i) a fixed-step choice $t = t_{\max}$, or (ii) an early-stopped choice $t = t_{\mathrm{stop}}^{(\ell)} \le t_{\max}$ where $t_{\mathrm{stop}}^{(\ell)}$ is the smallest full-step count achieving the target column-marginal tolerance in (4.11).

Layer-level errors used in Proposition 3.1. For a layer output $H_{\mathrm{res}}^{(\ell)} = S^{(2t)}$ (with t given by $t^{(\ell)}$ in (4.2)) define $\varepsilon_r^{(\ell)} := \|H_{\mathrm{res}}^{(\ell)}\mathbf{1} - \mathbf{1}\|_{\infty,\mathrm{vec}}$ and $\varepsilon_c^{(\ell)} := \|\mathbf{1}^\top H_{\mathrm{res}}^{(\ell)} - \mathbf{1}^\top\|_{\infty,\mathrm{vec}}$.

Output row sums. Every full column-then-row step ends with row normalization. Thus, for exact normalization and any $t \ge 1$, the output $H_{\mathrm{res}}^{(\ell)} = S^{(2t)}$ satisfies

$$H_{\mathrm{res}}^{(\ell)}\mathbf{1} = \mathbf{1}. \tag{4.8}$$

Strict positivity is preserved by both normalization maps, so

$$\|H_{\mathrm{res}}^{(\ell)}\|_\infty = 1. \tag{4.9}$$

Thus forward $A_{\max}$ mixer-path gain is controlled exactly at each layer. The remaining marginal quantity is the output column defect $\varepsilon_c^{(\ell)} = \|\mathbf{1}^\top H_{\mathrm{res}}^{(\ell)} - \mathbf{1}^\top\|_{\infty,\mathrm{vec}}$, which controls $\|H_{\mathrm{res}}^{(\ell)}\|_1$.

Early stopping to a target column tolerance. At each layer, we run full column–then–row Sinkhorn steps and measure the output column defect

$$\varepsilon_c^{(\ell)}(t) := \|\mathbf{1}^\top S^{(2t)} - \mathbf{1}^\top\|_{\infty,\mathrm{vec}}. \tag{4.10}$$

The layer stops at the first full-step count

$$t_{\mathrm{stop}}^{(\ell)} = \min\{1 \le t \le t_{\max} : \varepsilon_c^{(\ell)}(t) \le \bar{\varepsilon}_\ell\}, \tag{4.11}$$

if such a step exists, and otherwise returns the $t_{\max}$ output with a failed certificate for that layer. Appendix A summarizes the certificate evaluation. The column defect $\varepsilon_c^{(\ell)}(t)$ is always measured after a full column-then-row step, namely at $S^{(2t)}$.

Numerical normalizations. The theorems use exact division by current row and column sums. If code uses denominator stabilizers, floors, or mixed precision, the certificate should be computed from the realized output matrix, using its measured row and column defects. In the reported runs the stabilizer is 10^{-12}, so the measured row defect is negligible.

4.2 Dynamic range and Hilbert-metric control

Assume $S^{(0)}$ has bounded dynamic range: there exist $0 < m \le M$ such that $m \le S_{ij}^{(0)} \le M$ for all i, j, and define $\kappa := M/m$. If $S^{(0)} = \exp(\beta C)$, then

$$\kappa = \exp\big(\beta(\max_{i,j} C_{ij} - \min_{i,j} C_{ij})\big), \tag{4.12}$$

so clipping $C_{ij} \in [-R, R]$ implies $\kappa \le \exp(2\beta R)$.

Recall that for $u, v \in \mathbb{R}^n_{>0}$, the Hilbert projective metric is

$$d_H(u, v) := \log\Big(\max_i \frac{u_i}{v_i}\Big) - \log\Big(\min_i \frac{u_i}{v_i}\Big). \tag{4.13}$$

Initial projective spread. If $m \le S_{ij}^{(0)} \le M$ and $\kappa = M/m$, then the row-sum vector $u = S^{(0)}\mathbf{1}$ satisfies

$$d_H(u, \mathbf{1}) \le \log \kappa,$$

and the same bound holds for the column-sum vector because each row sum lies in $[nm, nM]$, so the ratio of its largest and smallest entries is at most κ.

Lemma 4.1 (Hilbert-to-marginal conversion)**.** *Let $u \in \mathbb{R}^n_{>0}$ satisfy $\frac{1}{n}\sum_{i=1}^n u_i = 1$ and set $\delta := d_H(u, \mathbf{1}) = \log\big(\max_i u_i / \min_i u_i\big)$. Then*

$$\|u - \mathbf{1}\|_{\infty,\mathrm{vec}} \le \Psi_n(\delta) := \frac{(n-1)(e^\delta - 1)}{(n-1) + e^\delta}. \tag{4.14}$$

Moreover, for any $\varepsilon \in [0, n-1)$,

$$\Psi_n(\delta) \le \varepsilon \quad \Longleftrightarrow \quad \delta \le \Psi_n^{-1}(\varepsilon) := \log\left(\frac{(n-1)(1+\varepsilon)}{(n-1) - \varepsilon}\right). \tag{4.15}$$

Proof. See Appendix B. □

Lemma 4.2 (Centered-log L^2 conversion to marginal error)**.** *Let $n \ge 2$, let $u \in \mathbb{R}^n_{>0}$ satisfy $n^{-1}\sum_i u_i = 1$, and put*

$$z := \Pi \log u, \qquad \eta := \|z\|_2.$$

Then

$$\|u - \mathbf{1}\|_{\infty,\mathrm{vec}} \le \Gamma_n(\eta), \tag{4.16}$$

where

$$\begin{aligned} \Gamma_n(\eta) &:= \max\{\Gamma_n^+(\eta), \Gamma_n^-(\eta)\}, \\ \Gamma_n^+(\eta) &:= \frac{n\exp(a)}{\exp(a) + (n-1)\exp(-a/(n-1))} - 1, \qquad a := \sqrt{\frac{n-1}{n}}\,\eta, \\ \Gamma_n^-(\eta) &:= \min\left\{1 - \exp(-\sqrt{2}\,\eta),\ 1 - \frac{\exp(-a)}{1 + \frac{e^a}{2n}\eta^2}\right\}. \end{aligned} \tag{4.17}$$

The upper-tail term is sharp: equality is attained when one centered log-coordinate equals a and the remaining $n-1$ centered log-coordinates equal $-a/(n-1)$. The lower-tail estimate is global and has the same optimal first-order coefficient. In particular, for small η,

$$\Gamma_n(\eta) = \sqrt{\frac{n-1}{n}}\,\eta + O(\eta^2). \tag{4.18}$$

Proof. See Appendix B. □

4.3 A finite-step sufficient rule

Classical Birkhoff–Bushell contraction theory for positive maps implies geometric contraction in the Hilbert metric with a rate controlled by the dynamic range (Birkhoff, 1957; Bushell, 1973). We specialize this contraction to the alternating normalization schedule.

Half-step indexing. We use half-step indices throughout the finite-step analysis. Starting from $S^{(0)} = A$, the alternating normalizations are

$$S^{(1)} = T_c(A), \qquad S^{(2)} = T_r T_c(A), \qquad S^{(3)} = T_c T_r T_c(A), \qquad \dots. \tag{4.19}$$

Thus $S^{(2t)}$ is the output after t full column-then-row steps. In exact arithmetic, $S^{(2t)}$ has exact row sums, whereas its column sums are generally only approximate. This is the origin of the one-half-step loss in the output column-error estimate: the column error at $S^{(2t)}$ is controlled at rate τ^{2t-1}, not τ^{2t}.

Birkhoff–Bushell contraction. For $A \in \mathbb{R}^{n\times n}_{>0}$, define the projective diameter and contraction factor by

$$\Delta(A) := \log \max_{i,k,j,\ell} \frac{A_{ij}A_{k\ell}}{A_{i\ell}A_{kj}}, \qquad \tau(A) := \tanh\left(\frac{\Delta(A)}{4}\right). \tag{4.20}$$

The Birkhoff–Bushell theorem (Birkhoff, 1957; Bushell, 1973) gives, for all $x, y \in \mathbb{R}^n_{>0}$,

$$d_H(Ax, Ay) \le \tau(A) d_H(x, y), \qquad d_H(A^\top x, A^\top y) \le \tau(A) d_H(x, y). \tag{4.21}$$

Componentwise inversion preserves Hilbert distance,

$$d_H(x^{-1}, y^{-1}) = d_H(x, y), \tag{4.22}$$

because it exchanges the maximum and minimum coordinate ratios.

Theorem 4.3 (Finite-step marginal-error bound)**.** *Let $A = S^{(0)} \in \mathbb{R}^{n\times n}_{>0}$, $n \ge 2$, and let $S^{(s)}$ be the alternating normalization iterates from Subsection 4.1. Set*

$$\delta_0 := d_H(S^{(1)}\mathbf{1}, \mathbf{1}), \qquad S^{(1)} = T_c(A).$$

Then, for every integer $t \ge 0$,

$$\varepsilon_r^{(2t+1)} := \|S^{(2t+1)}\mathbf{1} - \mathbf{1}\|_{\infty,\mathrm{vec}} \le \Psi_n\big(\delta_0 \tau(A)^{2t}\big). \tag{4.23}$$

For every integer $t \ge 1$, the full-step output satisfies

$$\varepsilon_c^{(2t)} := \|\mathbf{1}^\top S^{(2t)} - \mathbf{1}^\top\|_{\infty,\mathrm{vec}} \le \Psi_n\big(\delta_0 \tau(A)^{2t-1}\big). \tag{4.24}$$

Proof. Let $r^{(t)} \in \mathbb{R}^n_{>0}$ be the accumulated row-scaling vector at the beginning of the t-th full column-then-row step, with $r^{(0)} = \mathbf{1}$, modulo an irrelevant scalar factor. After the column normalization in full step t,

$$S^{(2t+1)} = \mathrm{diag}(r^{(t)}) A \,\mathrm{diag}\Big((A^\top r^{(t)})^{-1}\Big).$$

Its row-sum vector is therefore

$$u_t := S^{(2t+1)}\mathbf{1} = r^{(t)} \odot A(A^\top r^{(t)})^{-1}.$$

The following row normalization divides by this row-sum vector, thus

$$r^{(t+1)} = r^{(t)} \oslash u_t = \Big[A(A^\top r^{(t)})^{-1}\Big]^{-1} =: \Phi(r^{(t)}).$$

The map Φ contracts in Hilbert distance with coefficient $\tau(A)^2$. For $r, s \in \mathbb{R}^n_{>0}$, the Birkhoff–Bushell estimate (4.21) gives

$$\begin{aligned} d_H(A^\top r, A^\top s) &\le \tau(A) d_H(r, s), \\ d_H((A^\top r)^{-1}, (A^\top s)^{-1}) &= d_H(A^\top r, A^\top s), \\ d_H(A(A^\top r)^{-1}, A(A^\top s)^{-1}) &\le \tau(A) d_H((A^\top r)^{-1}, (A^\top s)^{-1}). \end{aligned}$$

Applying the inversion isometry once more gives

$$d_H(\Phi(r), \Phi(s)) \le \tau(A)^2 d_H(r, s).$$

Therefore,

$$d_H(r^{(t+1)}, r^{(t)}) \le \tau(A)^{2t} d_H(r^{(1)}, r^{(0)}).$$

Since $u_t = r^{(t)} \oslash r^{(t+1)}$,

$$d_H(u_t, \mathbf{1}) = d_H(r^{(t)}, r^{(t+1)}) \le \tau(A)^{2t}\delta_0, \qquad \delta_0 = d_H(u_0, \mathbf{1}) = d_H(S^{(1)}\mathbf{1}, \mathbf{1}).$$

After a column normalization the total mass is n, so $n^{-1}\sum_i (u_t)_i = 1$. Lemma 4.1 gives (4.23).

For the column sums at the full-step output $S^{(2t)}$, take $t \geq 1$, write

$$v_t := (\mathbf{1}^\top S^{(2t)})^\top.$$

Using the row-scaled representation before and after the preceding row normalization gives

$$v_t = (A^\top r^{(t)}) \oslash (A^\top r^{(t-1)}).$$

Therefore

$$d_H(v_t, \mathbf{1}) = d_H(A^\top r^{(t)}, A^\top r^{(t-1)}) \leq \tau(A) d_H(r^{(t)}, r^{(t-1)}) \leq \delta_0 \tau(A)^{2t-1}.$$

Since $S^{(2t)}$ has total mass n, Lemma 4.1 gives (4.24). □

Dynamic-range specialization. If $m \leq A_{ij} \leq M$ and $\kappa = M/m$, then $\Delta(A) \leq 2\log\kappa$, and therefore

$$\tau(A) \leq q := \frac{\kappa - 1}{\kappa + 1}.$$

For the first column-normalized iterate,

$$\frac{(S^{(1)}\mathbf{1})_i}{(S^{(1)}\mathbf{1})_k} \leq \max_j \frac{A_{ij}}{A_{kj}} \leq \kappa,$$

so $\delta_0 \leq \log\kappa$. Thus

$$\varepsilon_r^{(2t+1)} \leq \Psi_n\big((\log\kappa) q^{2t}\big), \qquad t \geq 0, \tag{4.25}$$

and

$$\varepsilon_c^{(2t)} \leq \Psi_n\big((\log\kappa) q^{2t-1}\big), \qquad t \geq 1. \tag{4.26}$$

For $\kappa > 1$, a sufficient step count for $\varepsilon_c^{(2t_{\max})} \leq \bar{\varepsilon}$ is

$$t_{\max} \geq \max\left\{1, \left\lceil \frac{1}{2}\left[1 + \frac{\log\big((\log\kappa)/\Psi_n^{-1}(\bar{\varepsilon})\big)}{\log(1/q)}\right]\right\rceil\right\}. \tag{4.27}$$

If A is already doubly stochastic, no normalization is needed; under the $t \geq 1$ convention one full step leaves it unchanged. If $\kappa = 1$, one full step is exact. The κ-only estimate is uniform over all matrices with the specified dynamic range. The matrix-specific bound in Theorem 4.3, and measured-error early stopping, can be substantially sharper.

Corollary 4.4 (Heterogeneous finite-step master certificate)**.** *For each layer ℓ, let $A_\ell > 0$ be normalized for $t_\ell \geq 1$ full column-then-row steps, and write*

$$H^{(\ell)} := S_\ell^{(2t_\ell)}, \qquad h_\ell := \Psi_n\big(\delta_{0,\ell}\tau(A_\ell)^{2t_\ell - 1}\big).$$

Let $c_\ell = (H^{(\ell)})^\top\mathbf{1} - \mathbf{1}$, $\delta_\ell = \delta_{\mathrm{D}}(H^{(\ell)})$, and $T_0 = I$. Then $\varepsilon_c^{(\ell)} \leq h_\ell$ and

$$\|T_L\|_1 \leq \min\left\{\prod_{\ell=1}^{L}(1+h_\ell),\ n,\ 1 + \sum_{\ell=1}^{L}\|T_{\ell-1}\|_1 h_\ell,\ 1 + \frac{1}{2}\sum_{\ell=1}^{L}\left(\prod_{j=1}^{\ell-1}\delta_j\right)\|c_\ell\|_{1,\mathrm{vec}}\right\}. \tag{4.28}$$

The four terms are respectively the product, structural, telescoping, and Dobrushin certificates. The final term also admits the fully marginal-error bound obtained by replacing $\frac{1}{2}\|c_\ell\|_{1,\mathrm{vec}}$ with $\lfloor n/2\rfloor h_\ell$.

Proof. Theorem 4.3 gives $\varepsilon_c^{(\ell)} \leq h_\ell$. Apply Proposition 3.1, Proposition 3.2, Lemma 3.3, and Proposition 3.4, then take the smallest of the four valid upper bounds. The marginal-error replacement follows from $\|c_\ell\|_{1,\mathrm{vec}} \leq 2\lfloor n/2\rfloor\varepsilon_c^{(\ell)}$. □

4.4 Matrix-specific local analysis

For a fixed matrix, the nonlinear Sinkhorn map in logarithmic projective coordinates gives matrix-specific information beyond the global dynamic-range bound. Its derivative at the Sinkhorn limit gives the asymptotic rate $\sigma_2(P)^2$. A finite-orbit derivative-envelope certificate is recorded in Appendix B.1.

Let

$$P = \operatorname{diag}(r_*) A \operatorname{diag}(c_*) \in \mathcal{B}_n$$

be the limiting doubly stochastic scaling of $A \in \mathbb{R}^{n\times n}_{>0}$, which exists and is unique, with r_*, c_* unique up to reciprocal scaling, by the Sinkhorn–Knopp theorem for strictly positive matrices (Sinkhorn and Knopp, 1967). We work on $\mathbf{1}^\perp$ and define

$$\mathcal{F}(x) := \Pi\left[-\log\left(P\left((P^\top e^x)^{-1}\right)\right)\right], \qquad \mathcal{L}(x) := \Pi \log(P^\top e^x), \tag{4.29}$$

where all exponentials, logarithms, and inverses are componentwise. If

$$x_t := \Pi \log \frac{r^{(t)}}{r_*},$$

then

$$x_{t+1} = \mathcal{F}(x_t), \tag{4.30}$$

and the centered logarithmic output column error after $t \geq 1$ full steps is

$$z_t := \Pi \log v_t = \mathcal{L}(x_t) - \mathcal{L}(x_{t-1}), \qquad v_t := (\mathbf{1}^\top S^{(2t)})^\top. \tag{4.31}$$

Set

$$K := PP^\top, \qquad \rho_* := \|K\Pi\|_{2\to 2}.$$

Lemma 4.5 (Local linearization)**.** *With the notation above,*

$$\rho_* = \sigma_2(P)^2, \qquad D\mathcal{F}(0) = K|_{\mathbf{1}^\perp}. \tag{4.32}$$

In particular,

$$x_{t+1} = Kx_t + O(\|x_t\|_2^2) \tag{4.33}$$

locally. For every $\eta > 0$, after restricting to a sufficiently small neighborhood of the origin,

$$\|x_t\|_2 \leq C_\eta(\rho_* + \eta)^t \|x_0\|_2, \tag{4.34}$$

and the centered logarithmic column error satisfies

$$z_t = \Pi P^\top (K - I) K^{t-1} x_0 + O\big((\rho_* + \eta)^t \|x_0\|_2^2\big)\,. \tag{4.35}$$

Proof. The recurrence (4.30) follows from

$$A^\top r^{(t)} = c_*^{-1} \odot P^\top e^{x_t}, \qquad (A^\top r^{(t)})^{-1} = c_* \odot (P^\top e^{x_t})^{-1},$$

where the scalar ambiguity in e^{x_t} is removed by Π. The row-scaling update gives

$$e^{x_{t+1}} \sim \left[P\left((P^\top e^{x_t})^{-1}\right)\right]^{-1},$$

which is (4.30). Since $P^\top \mathbf{1} = \mathbf{1}$,

$$P^\top e^x = \mathbf{1} + P^\top x + O(\|x\|_2^2), \qquad (P^\top e^x)^{-1} = \mathbf{1} - P^\top x + O(\|x\|_2^2).$$

Applying P, taking $-\log$, and projecting gives $\mathcal{F}(x) = \Pi P P^\top x + O(\|x\|_2^2)$. On $\mathbf{1}^\perp$, $K = PP^\top$ is symmetric and has operator norm $\sigma_2(P)^2$, proving (4.32) and (4.33). The estimate (4.34) follows from the standard local stability estimate for a C^2 map at an attracting fixed point. Since $D\mathcal{L}(0) = \Pi P^\top$, Equation (4.31) gives (4.35). □

Proposition 4.6 (Asymptotic local column-error rate)**.** *Assume* $0 < \rho_* < 1$ *and take* x_0 *sufficiently close to the origin. There is* $w_* \in \ker(K - \rho_* I)$ *such that*

$$x_t = \rho_*^t w_* + o(\rho_*^t). \tag{4.36}$$

If $w_* \neq 0$*, then*

$$z_t = (\rho_* - 1)\rho_*^{t-1} \Pi P^\top w_* + o(\rho_*^{t-1}), \tag{4.37}$$

and

$$\frac{\|z_{t+1}\|_2}{\|z_t\|_2} \longrightarrow \rho_* = \sigma_2(P)^2. \tag{4.38}$$

Proof. Write

$$x_{t+1} = K x_t + N(x_t), \qquad \|N(x)\|_2 \le C\|x\|_2^2$$

in a sufficiently small neighborhood of the origin. Choose α with $\rho_* < \alpha < \sqrt{\rho_*}$. After shrinking the neighborhood if necessary, Lemma 4.5 gives $\|x_t\|_2 = O(\alpha^t)$, so $N(x_t) = O(\alpha^{2t}) = o(\rho_*^t)$. Let E_* be the eigenspace of the symmetric matrix K corresponding to ρ_*, and let Q_* be the orthogonal projection onto E_*. The variation-of-constants formula gives

$$\rho_*^{-t} Q_* x_t = Q_* x_0 + \sum_{s=0}^{t-1} \rho_*^{-s-1} Q_* N(x_s).$$

The series converges because $\alpha^2 < \rho_*$, and its limit defines $w_* \in E_*$. On $E_*^\perp \cap \mathbf{1}^\perp$, all eigenvalues of K are strictly smaller than ρ_*, so the complementary component is $o(\rho_*^t)$. This proves (4.36).

Using $D\mathcal{L}(0) = \Pi P^\top$,

$$\begin{aligned} z_t &= \mathcal{L}(x_t) - \mathcal{L}(x_{t-1}) \\ &= \Pi P^\top (x_t - x_{t-1}) + O(\|x_t\|_2^2 + \|x_{t-1}\|_2^2), \end{aligned}$$

which gives (4.37). If $w_* \neq 0$, then

$$\|P^\top w_*\|_2^2 = w_*^\top P P^\top w_* = \rho_* \|w_*\|_2^2 > 0,$$

so the leading coefficient in (4.37) is nonzero and (4.38) follows. □

If $\rho_* = 0$, the linear term vanishes and the local decay is determined by quadratic or higher-order terms.

Spectral gap. For strictly positive A the Sinkhorn limit P is strictly positive. Its singular values satisfy $\sigma_2(P) < \sigma_1(P) = 1$, so $\rho_* = \sigma_2(P)^2 < 1$ and the local recursion is attracting. When $0 < \rho_* < 1$, the ratio in (4.38) holds whenever the nonlinear leading coefficient w_* is nonzero. If that coefficient vanishes, the decay is governed by the largest lower mode, or by higher-order terms in the degenerate case $\rho_* = 0$.

4.5 Certificates versus predictors

The global κ-only and matrix-adaptive Hilbert estimates are finite-step certificates. By contrast, the spectral rate $\rho_* = \sigma_2(P)^2$ and the finite-step expansions below are predictors used to interpret observed decay curves. Appendix B.1 records a conditional derivative-envelope certificate on a verified neighborhood of the Sinkhorn limit.

For the plots we use the first-order local predictor

$$\ell_t^{[1]} = \Pi P^\top (K - I) K^{t-1} x_0, \qquad K = PP^\top, \tag{4.39}$$

followed by exponentiation and recentering,

$$\widehat{v}_t^{[1]} = \frac{n \exp(\ell_t^{[1]})}{\sum_{j=1}^n \exp((\ell_t^{[1]})_j)}, \qquad \widehat{\varepsilon}_{c,1}^{(2t)} = \|\widehat{v}_t^{[1]} - \mathbf{1}\|_{\infty,\mathrm{vec}}. \tag{4.40}$$

The second-order predictor is given in Appendix C; both predictors are used below to interpret the measured finite-step decay; neither enters the deterministic budget certificate.

Dynamic range and routing selectivity. Smaller dynamic range κ, obtained for instance by clipping or smaller temperature β, improves the contraction factor and reduces the step budget needed for a target marginal tolerance. At the same time, as $\kappa \downarrow 1$, the feasible matrices concentrate near $n^{-1}\mathbf{1}\mathbf{1}^\top$, reducing selective routing. Thus κ trades off certificate strength against routing expressivity.

4.6 Mixer-path calibration from $A_{\max}$ budgets

Combining the depth-wise product certificate (Proposition 3.1) with the finite-step Sinkhorn bound (Theorem 4.3) gives a direct calibration rule from target $A_{\max}$ budgets to implementable parameters (logit dynamic range and step count).

Theorem 4.7 (Mixer-path calibration rule: $(\kappa, t_{\max}) \Rightarrow \|T_L\|_1$ budget)**.** *Assume each mixer $H_{\mathrm{res}}^{(\ell)}$ is entrywise nonnegative and produced by exponentiating logits with dynamic range at most κ, followed by $t_{\max} \geq 1$ ideal exact full Sinkhorn steps in the column–then–row order. Suppose the desired per-layer column-marginal tolerance is ε_c. If*

$$\Psi_n\big((\log \kappa) q^{2t_{\max}-1}\big) \leq \varepsilon_c, \qquad q = \frac{\kappa - 1}{\kappa + 1}, \tag{4.41}$$

then every output mixer satisfies $\varepsilon_c^{(\ell)} \leq \varepsilon_c$ and the backward $A_{\max}$ gain of the depth-L mixer product T_L satisfies

$$\|T_L\|_1 \leq (1 + \varepsilon_c)^L. \tag{4.42}$$

In particular, to enforce a backward budget $\|T_L\|_1 \leq B_1$ it is sufficient to choose

$$\varepsilon_c \leq B_1^{1/L} - 1 \;\approx\; \frac{\log B_1}{L} \qquad (\textit{for small } \varepsilon_c), \tag{4.43}$$

and then choose $t_{\max}$ satisfying (4.41). For $\kappa > 1$, an explicit sufficient choice for a matrix that is not already doubly stochastic is

$$t_{\max} \geq \max\left\{1, \left\lceil \frac{1}{2}\left[1 + \frac{\log\big((\log \kappa)/\Psi_n^{-1}(\varepsilon_c)\big)}{\log(1/q)}\right]\right\rceil\right\}. \tag{4.44}$$

For $\kappa = 1$, one full step is exact.

Proof. The column-marginal estimate (4.26) gives $\varepsilon_c^{(\ell)} \leq \varepsilon_c$ under (4.41). Proposition 3.1 then gives

$$\|T_L\|_1 \leq \prod_{\ell=1}^{L} (1 + \varepsilon_c^{(\ell)}) \leq (1 + \varepsilon_c)^L.$$

The tolerance choice (4.43) is exactly the condition $(1 + \varepsilon_c)^L \leq B_1$. Solving (4.41) for $t_{\max}$ gives (4.44). □

Budgets at or above the structural ceiling. By Proposition 3.2, in the ideal column–then–row order one always has $\|T_L\|_1 \leq n$ after a single full step, with no dependence on the dynamic range. Thus the finite-step budget (4.41) is needed only when the target satisfies $B_1 < n$; for $B_1 \geq n$ the structural ceiling already certifies the budget and $t_{\max} = 1$ suffices. Equivalently, the per-layer tolerance (4.43) need only be enforced down to the point where $\prod_\ell (1 + \varepsilon_c^{(\ell)}) = n$, after which the ceiling is the binding constraint. The telescoping bound (3.8) can be used in place of the product certificate to certify the same target with larger per-layer tolerances whenever the realized prefix gains are sub-multiplicative.

Choosing the target budget. The quantity B_1 is an allowed worst-case amplification through the mixing pathway, not a universally optimal training hyperparameter. For example, $B_1 = 1.066075 = (1 + 5 \times 10^{-4})^{128}$ permits at most a 6.6% certified backward $A_{\max}$ amplification from the depth-128 mixer product under the uniform allocation. A value close to one is appropriate when the mixing path is intended to be nearly nonexpansive or when most of a full-block gain budget is reserved for the nonlinear branches. A larger target below the structural ceiling n relaxes the per-layer tolerance and can reduce normalization work. Thus B_1 should be chosen as an application-level bound on mixer-path amplification: smaller values impose tighter normalization requirements, while larger values reduce normalization cost.

4.7 Adaptive global-budget early stopping

The uniform rule $\varepsilon_c \le B_1^{1/L} - 1$ is certificate-valid but not step-efficient: some layers reach a much smaller error than required, while other layers need more iterations. The alternative is to allocate the logarithmic product budget online.

Lemma 4.8 (Adaptive log-budget early stopping)**.** *Fix a target backward budget $B_1 \ge 1$ and set $b_1 = \log B_1$. Suppose the layers are processed in order. Define $s_0 = 0$, and before layer ℓ set*

$$\bar{\varepsilon}_\ell := \exp\left(\frac{b_1 - s_{\ell-1}}{L - \ell + 1}\right) - 1. \tag{4.45}$$

Run finite-step Sinkhorn at layer ℓ until

$$\varepsilon_c^{(\ell)} \le \bar{\varepsilon}_\ell, \tag{4.46}$$

and then update

$$s_\ell = s_{\ell-1} + \log(1 + \varepsilon_c^{(\ell)}). \tag{4.47}$$

If every layer reaches its assigned tolerance before the step cap, then

$$\sum_{\ell=1}^{L} \log(1 + \varepsilon_c^{(\ell)}) \le b_1, \qquad \|T_L\|_1 \le B_1. \tag{4.48}$$

Proof. Let $r_{\ell-1} := b_1 - s_{\ell-1}$ be the remaining log-budget. By construction,

$$\log(1 + \bar{\varepsilon}_\ell) = \frac{r_{\ell-1}}{L - \ell + 1}.$$

Since $\varepsilon_c^{(\ell)} \le \bar{\varepsilon}_\ell$,

$$s_\ell \le s_{\ell-1} + \frac{b_1 - s_{\ell-1}}{L - \ell + 1}.$$

Equivalently,

$$b_1 - s_\ell \ge \frac{L - \ell}{L - \ell + 1}(b_1 - s_{\ell-1}).$$

Induction gives $s_L \le b_1$. Proposition 3.1 then implies $\|T_L\|_1 \le \exp(s_L) \le B_1$. □

Certificate failure under a finite cap. Lemma 4.8 applies when every layer reaches its allocated tolerance before the step cap. If some layer does not reach that tolerance by $t_{\max}$, the resulting stack is not certified by this stopping rule to satisfy $\|T_L\|_1 \le B_1$. One may then increase $t_{\max}$, reduce the logit dynamic range, or choose a larger budget.

For implementation, the rule is evaluated sequentially. After observing

$$s_{\ell-1} = \sum_{j<\ell} \log(1 + \varepsilon_c^{(j)}),$$

layer ℓ is assigned the tolerance

$$\bar{\varepsilon}_\ell = \exp\left(\frac{\log B_1 - s_{\ell-1}}{L - \ell + 1}\right) - 1. \tag{4.49}$$

The layer then uses the early-stopping rule (4.11). If the cap is reached before $\varepsilon_c^{(\ell)} \le \bar{\varepsilon}_\ell$, the certificate fails for that layer; otherwise $s_\ell = s_{\ell-1} + \log(1 + \varepsilon_c^{(\ell)})$. A compact implementation summary is given in Appendix A.

4.8 Temperature–clipping calibration

The dynamic range is controlled in implementations through logit clipping and temperature. Suppose

$$C_{ij} \in [-R, R], \qquad A_{ij} = \exp(\beta C_{ij}). \tag{4.50}$$

Then

$$\kappa(A) \le \exp(2\beta R), \qquad \frac{\kappa(A)-1}{\kappa(A)+1} \le \tanh(\beta R). \tag{4.51}$$

Therefore the conservative finite-step rule becomes

$$\varepsilon_c^{(2t)} \le \Psi_n\big(2\beta R \tanh(\beta R)^{2t-1}\big). \tag{4.52}$$

Fix n, a full-step budget $t_{\max} \ge 1$, and a target column tolerance $\bar{\varepsilon}$. If

$$\Psi_n\big(2\beta R \tanh(\beta R)^{2t_{\max}-1}\big) \le \bar{\varepsilon}, \tag{4.53}$$

then every clipped logit matrix $C \in [-R, R]^{n\times n}$ produces, after $t_{\max}$ full column-then-row Sinkhorn steps, a nonnegative mixer satisfying

$$\varepsilon_c^{(2t_{\max})} \le \bar{\varepsilon}.$$

This follows immediately from (4.51), the monotonicity of Ψ_n, and Theorem 4.3. A depth-L backward budget B_1 is certified whenever $\bar{\varepsilon} \le B_1^{1/L} - 1$.

Theorem 4.7 converts a backward mixer-path budget into a column-marginal tolerance and a sufficient Sinkhorn step count. Table 1 gives conservative examples.

Table 1: Conservative full-step budgets $t_{\max}$ sufficient to achieve $\Psi_n((\log\kappa)q^{2t_{\max}-1}) \le \varepsilon_{\mathrm{tol}}$ (Theorem 4.3). Entries are sufficient $t_{\max}$ values.

κ	$\varepsilon_{\mathrm{tol}}$	$n=4$	$n=8$	$n=16$
2	5×10^{-3}	3	3	3
2	1×10^{-3}	4	4	4
2	5×10^{-4}	4	4	4
5	5×10^{-3}	8	8	8
5	1×10^{-3}	10	10	10
5	5×10^{-4}	11	11	11
10	5×10^{-3}	16	16	16
10	1×10^{-3}	20	20	20
10	5×10^{-4}	21	22	22
20	5×10^{-3}	32	32	33
20	1×10^{-3}	40	40	41
20	5×10^{-4}	43	44	44

A full Sinkhorn step (one column and one row normalization) costs $O(n^2)$ time and $O(n^2)$ memory for dense $n \times n$ mixers. Smaller tolerances or larger dynamic range require larger $t_{\max}$ and greater per-layer cost.

The calibration is motivated by the $A_{\max}$ behavior reported in Xie et al. (2025): single-layer gains can remain moderate while the composite backward gain grows substantially with depth. Finite-step Sinkhorn normalization gives a deterministic way to prevent this within the mixer pathway: once the measured marginal errors meet the prescribed tolerances, the corresponding depth-wise product bound follows.

5 Numerical diagnostics

The experiments evaluate certificate behavior and finite-step calibration under controlled conditions. We report marginal errors, the product certificate $\prod_{\ell=1}^{L}(1+\varepsilon_c^{(\ell)})$, the exact backward mixing gain $\|T_L\|_1$, cap failures, normalization cost, and local convergence rates.

We use two CIFAR-10 configurations: a matched transformer run and a depth-128 stress run. Within each paired comparison, the softmax and Sinkhorn models use identical augmented minibatches, matched non-mixer initialization, and a deterministic per-epoch data order. We record the mixer matrices after the same normalization used in training. Marginal errors are measured as $\varepsilon_c^{(\ell)} = \|\mathbf{1}^\top H^{(\ell)} - \mathbf{1}^\top\|_{\infty,\mathrm{vec}}$ and $\varepsilon_r^{(\ell)} = \|H^{(\ell)}\mathbf{1} - \mathbf{1}\|_{\infty,\mathrm{vec}}$. The exact realized backward mixing gain is computed by the vector recursion $w_L = \mathbf{1}$, $w_{\ell-1} = H^{(\ell)\top} w_\ell$, and $\|T_L\|_1 = \|w_0\|_\infty$, as in Equation (3.4). The same final-epoch weights are also evaluated with fixed $t = t_{\max}$ to separate early-stop behavior from what would be obtained by spending the whole cap.

5.1 Matched transformer run

We run a short-horizon CIFAR-10 transformer-style classifier with $n = 8$ stream copies and trainable stream mixers in each block. The run uses 5000 training images, 2000 test images, eight epochs, and depth $L = 24$. Across seeds $0, 1, 2$, the two rows in Table 2 use identical initial weights, minibatches, optimizer settings, and architecture within each seed; only the stream-mixer normalization changes. The row-only softmax mixer is row-stochastic but has no column control, while the Sinkhorn row uses the adaptive log-budget controller.

The row-only run reaches an exact depth-wise backward mixer gain close to the structural ceiling $n = 8$, whereas the Sinkhorn-budget run keeps $\|T_L\|_1$ near one with comparable task loss and accuracy. Figure 1 shows the same quantities over training. The row-only model has marginally higher mean test accuracy. The large reduction in mixer-path gain therefore does not translate into a corresponding accuracy improvement in this short-horizon experiment; the certificate controls a specific worst-case amplification mechanism rather than serving as an accuracy objective. As an additional training diagnostic, the final-epoch mean full-network gradient norms are 1.213 ± 0.051 for row-only softmax and 1.253 ± 0.038 for the Sinkhorn controller (mean±std over seeds). The per-seed trajectory maxima over all eight epochs are 5.32 ± 2.18 and 4.99 ± 0.91, respectively, and no nonfinite update occurs in either policy.

Policy	Acc.	Loss	Exact $\|T_L\|_1$	Product cert.
row-only softmax	0.350±0.026	1.784±0.045	7.981±0.004	4.46e+21±1.72e+18
Sinkhorn budget	0.339±0.033	1.796±0.049	1.000±2.82e-06	1.006±0.002

Table 2: Matched short-horizon CIFAR-10 transformer diagnostic. Both rows use the same initial weights and minibatches within each seed; only the stream-mixer normalization changes. The row-only mixer is row-stochastic but not column-controlled, while the Sinkhorn row uses the adaptive log-budget controller. Statistics are mean±std over seeds 0, 1, 2.

Table 3 compares the available backward-gain certificates on the saved final mixer stacks. For row-only softmax, the refined Dobrushin certificate is 7.9868 ± 0.0002, slightly sharper than the structural ceiling $n = 8$ and close to the exact gain 7.9814 ± 0.0036; the product and telescoping bounds are much looser. For the Sinkhorn-controlled stacks, the Dobrushin certificate is 1.000079 ± 0.000009, compared with the exact gain 1.000015 ± 0.000002. Among the reported bounds, the layerwise finite-horizon certificate is the smallest in both regimes.

Mixer policy	Product	Structural	Telescoping	Dobrushin	Exact
row-only softmax	$(4.459 \pm 0.001) \times 10^{21}$	8	1289.47 ± 0.57	7.98679 ± 0.00015	7.9814 ± 0.0036
Sinkhorn budget	1.0063 ± 0.0016	8	1.0063 ± 0.0016	1.000079 ± 0.000009	1.000015 ± 0.000002

Table 3: Comparison of backward-gain certificates on the saved final matched-run mixer stacks. Statistics are mean±std over three seeds.

The remaining diagnostics for the matched transformer run are reported in Appendix A.2. The column-marginal errors confirm that the Sinkhorn controller enforces the intended finite-step tolerance, while the

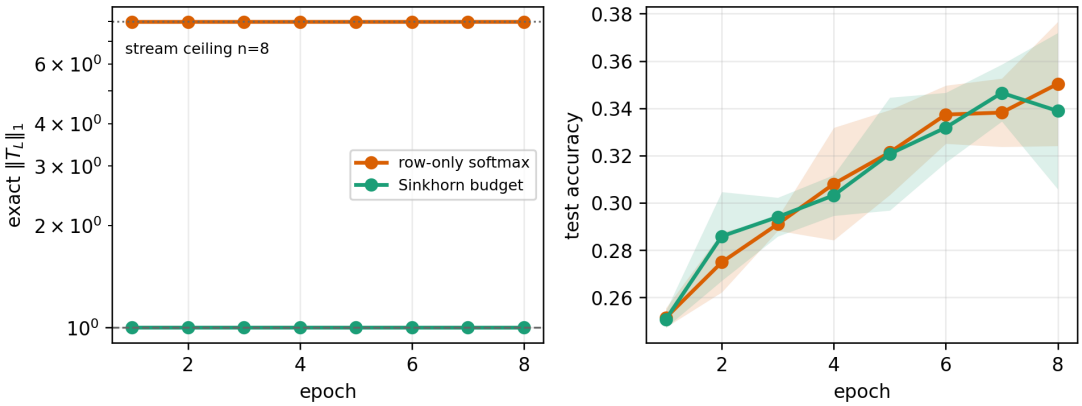


Figure 1: Matched short-horizon transformer run on a CIFAR-10 subset. Row-only softmax mixers nearly saturate the structural backward-gain ceiling $n = 8$, while the column-then-row Sinkhorn controller keeps the exact propagated mixer gain near one. Test accuracy remains comparable under the same training setup.

Dobrushin audit compares the refined finite-horizon and depth-uniform controls on the saved final mixer stacks.

5.2 Controller ablation and finite-step diagnostics

The main CIFAR-10 stress setting uses $L = 128$, $n = 8$ streams, feature dimension $d = 128$, SGD with learning rate 0.02, momentum 0.9, weight decay 5×10^{-4}, Sinkhorn tolerance $\varepsilon_{\text{tol}} = 5 \times 10^{-4}$, and cap $t_{\max} = 80$. The softmax branch uses the same architecture but a row-stochastic softmax mixer. This stress setting uses a larger mixer-logit scale and temperature than the matched run above, making early stopping, cap failures, and dynamic-range effects more visible.

The training run records final-epoch dynamic range, early-stopping counts, exact propagated gain, product certificates, local spectral rates, and cap failures. Additional audit tables and raw logs are included in the supplementary archive.

For the certificate-aware ablation, define

$$\mathcal{R}_{\text{budget}} = \left[\sum_{\ell=1}^{L} \log(1 + \varepsilon_c^{(\ell)}) - \log B_1 \right]_+^2$$

and train with

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda_{\text{cert}} \mathcal{R}_{\text{budget}} + \lambda_\kappa \sum_{\ell=1}^{L} [\log \kappa_\ell - \log \kappa_{\max}]_+^2 .$$

This is an optimization penalty rather than a substitute for certification: the hard mixer-product bound is still evaluated from the realized post-normalization defects.

To test the controller in Lemma 4.8, we reran the same CIFAR-10 stress setting with only the Sinkhorn stopping/control policy changed. Table 4 compares fixed cap, uniform early stopping, adaptive log-budget stopping, and an adaptive policy trained with the penalty above.

The fixed-cap policy spends substantially more Sinkhorn steps and produces a certificate well below the prescribed budget. Uniform tolerance gives similar accuracy and exact realized backward gain using about one fifth of the Sinkhorn iterations. The adaptive log-budget policy spends the intended product budget more directly and slightly reduces iterations and epoch time relative to uniform tolerance. The certificate-aware penalty achieves similar accuracy and certified gain but incurs substantially higher training cost in this implementation (570.2 s/epoch versus 309.6 s/epoch for adaptive stopping), reflecting the additional

differentiable certificate computation. Among the tested policies that retain a hard realized certificate, adaptive stopping has the lowest reported training cost.

Policy	Acc.	Time/epoch	Avg. steps	Cert.	Exact $\|T_L\|_1$	Cap fail.
Fixed $t = 80$	0.584 ± 0.037	758.0 ± 6.1s	80.00 ± 0.00	1.0004 ± 0.0003	1.00000 ± 0.00000	$0.00 \pm 0.00\%$
Uniform tol.	0.584 ± 0.037	325.4 ± 17.6s	16.36 ± 0.71	1.0521 ± 0.0012	1.00029 ± 0.00014	$0.26 \pm 0.37\%$
Adaptive budget	0.581 ± 0.034	309.6 ± 19.1s	15.86 ± 0.73	1.0659 ± 0.0001	1.00029 ± 0.00014	$0.26 \pm 0.37\%$
Cert.-aware penalty	0.584 ± 0.038	570.2 ± 34.2s	15.80 ± 0.72	1.0656 ± 0.0000	1.00030 ± 0.00014	$0.26 \pm 0.37\%$

Table 4: Focused controller ablation under the CIFAR-10 stress setting. The adaptive and certificate-aware policies use the shared budget $B_1 = (1 + 5 \cdot 10^{-4})^{128} = 1.066075$. Statistics are computed over seeds $0, 1, 2$; the associated local rates are all approximately $\rho_* = 0.62$.

Sensitivity to the target budget. To isolate the certificate–compute trade-off from retraining, we apply the adaptive controller post hoc to the three saved final row-only mixer stacks from the matched transformer run, treating each positive row-stochastic mixer as the starting matrix for column-then-row normalization. Table 5 varies only B_1; it does not evaluate task accuracy after renormalization. Relaxing the budget from 1.005 to 1.5 reduces the mean full-step count from 2.01 to 1.00 and the measured CPU normalization time from about 0.30 to 0.17 ms per 24-layer stack. Every realized product certificate remains below its assigned budget, every cap-failure rate is zero, and the exact gain remains between 1.0000 and 1.0062. At looser budgets the certificate can remain well below the assigned value because full Sinkhorn steps are discrete; B_1 is only an upper bound.

Budget B_1	Avg. steps	CPU ms/stack	Product cert.	Exact $\|T_L\|_1$	Cap fail.
1.005	2.0139 ± 0.0196	0.3044 ± 0.0062	1.0013 ± 0.0002	1.0000 ± 0.0000	0%
1.01207	1.9583 ± 0.0000	0.2937 ± 0.0014	1.0063 ± 0.0017	1.0001 ± 0.0001	0%
1.03	1.8333 ± 0.0589	0.2771 ± 0.0087	1.0225 ± 0.0052	1.0001 ± 0.0001	0%
1.06608	1.6667 ± 0.0680	0.2588 ± 0.0052	1.0427 ± 0.0067	1.0001 ± 0.0001	0%
1.1	1.4444 ± 0.1094	0.2242 ± 0.0147	1.0742 ± 0.0109	1.0001 ± 0.0001	0%
1.25	1.0556 ± 0.0520	0.1754 ± 0.0070	1.1584 ± 0.0101	1.0038 ± 0.0025	0%
1.5	1.0000 ± 0.0000	0.1647 ± 0.0003	1.1776 ± 0.0284	1.0061 ± 0.0008	0%

Table 5: Post-hoc global-budget sensitivity on saved final matched-run mixer stacks. Timing is for normalization of one 24-layer stack on a single CPU process and excludes file loading. Statistics are mean±std over three seeds.

5.3 Calibration and local-predictor diagnostics

Figure 2 tests the design inequality (4.52) on random clipped logit matrices with $n = 8$. The measured column errors are always below the finite-precision certificate when the certificate is above the numerical floor. The grid also shows why the temperature–clipping rule is a conservative design certificate rather than a predictor: for large βR, the worst-case certificate is nearly vacuous even though the measured errors decrease rapidly with $t_{\max}$.

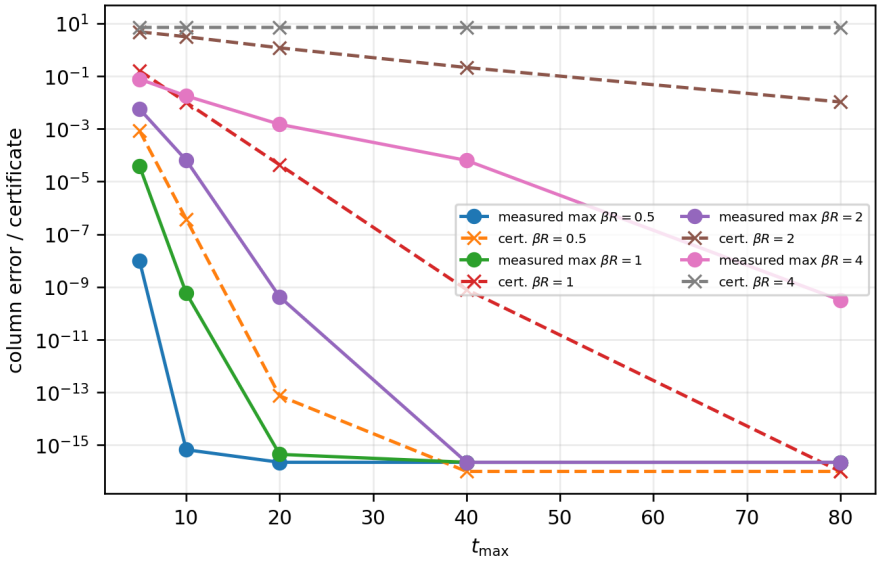


Figure 2: Temperature–clipping diagnostic for the conservative design rule (4.52). Solid curves are measured maximum column errors over random clipped logits; dashed curves are the corresponding worst-case certificates.

For each Sinkhorn layer, we compute the fixed-$t_{\max}$ matrix A^{fix}, use it as the realized doubly stochastic approximation P, and report $\rho_* = \sigma_2(P)^2$. We also report the empirical final-step ratio $\varepsilon_c^{(2t+2)}/\varepsilon_c^{(2t)}$ when the denominator is numerically meaningful. This ratio is diagnostic only; near the early-stopping tolerance or numerical floor it need not estimate the asymptotic local rate.

For trained adaptive Sinkhorn layers, the local first- and second-order predictors track the measured column-error decay much more closely than the global Hilbert certificates. The Hilbert bounds remain conservative, while the local expansions follow the measured decay. Figure 3 shows a representative trained layer.

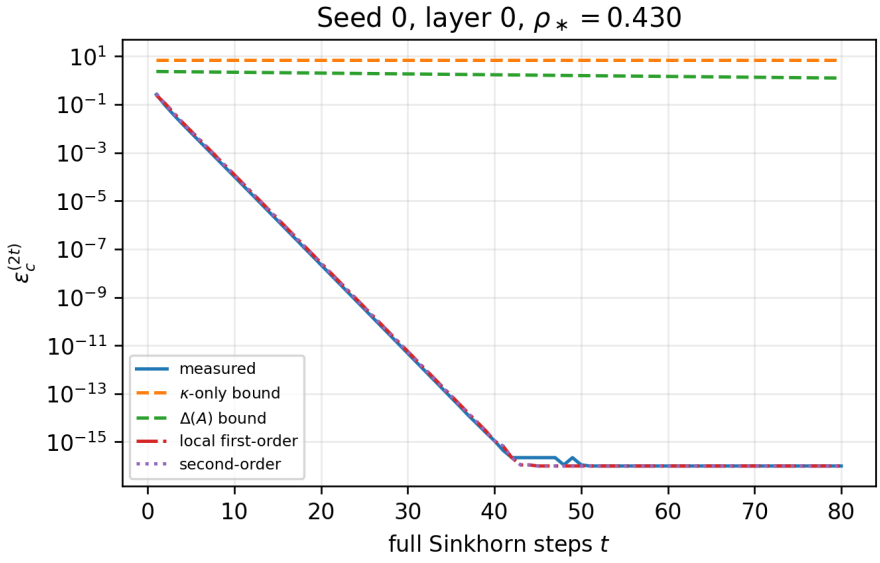


Figure 3: Representative local-predictor comparison on a trained adaptive Sinkhorn layer. The local first- and second-order predictors track the measured Sinkhorn decay closely, while the Hilbert certificates remain conservative.

The stress run operates in a high-dynamic-range regime; the corresponding final-epoch dynamic-range audit is reported in Appendix A.

Appendix A.3 contains a compact stress-run audit figure computed from the same logs.

6 Discussion and practical guidance

6.1 Practical guidance

The certificates are most informative when the desired backward mixer-path budget lies below the structural ceiling n. For nonnegative row-stochastic mixers, $\|T_L\|_1 \le n$ independently of depth and of the Sinkhorn schedule. Measured column defects provide the multiplicative certificate, while the telescoping bound can sharpen it by incorporating realized prefix gains. When the mixers are uniformly scrambling, the Dobrushin estimate gives a complementary depth-uniform audit. In the matched row-only run, its finite-horizon value is only slightly below the structural ceiling; for the Sinkhorn-controlled mixers it is close to one.

For a target $B_1 < n$, the small-error relation $\sum_{\ell=1}^{L} \varepsilon_c^{(\ell)} \approx \log B_1$ gives a useful scale for allocating layerwise tolerances. Theorem 4.7 converts those tolerances into conservative requirements on $(\kappa, t_{\max})$, or equivalently on temperature, clipping, and the step cap. The sensitivity study in Table 5 shows the corresponding compute trade-off directly: tighter budgets require more normalization work, while the realized exact gain can remain much smaller than the certificate.

The global Hilbert estimate should be read as a design certificate rather than a prediction of the observed convergence rate. Matrix-specific spectral quantities and the first- and second-order expansions are useful diagnostics once a realized matrix is available. Adaptive stopping uses measured marginal errors to reduce computation, and Equation (3.4) gives the exact realized backward gain without forming the full product.

6.2 Limitations

Worst-case and finite-step bounds. The product and Hilbert-metric bounds are sufficient worst-case certificates and can be conservative. The structural ceiling, telescoping refinement, matrix-specific local analysis, and exact propagated gain provide progressively more realized information when the corresponding quantities are available. The local spectral and second-order formulas describe finite-step decay near the Sinkhorn limit; they are predictors, not uniform certificates.

Nonlinear blocks and training dynamics. The nonlinear block F enters only through Jacobian norm bounds such as Proposition C.2 and Corollary C.3. We do not model how training shapes $\widetilde{H_{\mathrm{res}}}^{(\ell)}$ or how the certificates interact with optimization, data distribution, or generalization.

Architecture and implementation. The statements use streams-as-rows and the column–then–row Sinkhorn order. Streams-as-columns, damping, mixed precision, sparse mixers, and masked supports require explicit translation and can change which marginal error remains at the output. Exact doubly stochastic parameterizations such as mHC-lite avoid finite-step marginal error altogether; the present calibration is relevant when iterative Sinkhorn normalization is retained. We compare these approaches conceptually but do not benchmark exact parameterizations against iterative Sinkhorn in this study.

6.3 Future work

Open directions include inexpensive certification of the local derivative envelopes in Appendix B.1, sharper support- or gap-dependent refinements of the Dobrushin certificate, broader controller and certificate-penalty ablations across datasets and architectures, and full-network bounds that couple mixer certificates to nonlinear-branch Jacobians.

7 Conclusion

We developed deterministic finite-step bounds for nonnegative Sinkhorn-normalized stream mixers. Marginal errors translate directly into forward and backward induced-norm control, while Hilbert-projective contraction connects logit dynamic range and Sinkhorn step count to a depth-wise mixer budget. The structural ceiling $\|T_L\|_1 \leq n$ and the telescoping refinement provide complementary worst-case and realized controls, while the matrix-specific local analysis describes the observed Sinkhorn convergence rate near the limit.

The same framework gives practical calibration and adaptive stopping rules. In the focused ablations, adaptive log-budget stopping reaches the same mixer-path targets as a uniform tolerance while using fewer Sinkhorn steps, whereas the certificate-aware penalty is more expensive in the present implementation. In the matched transformer diagnostic, Sinkhorn normalization reduces the exact backward mixer gain from near the structural ceiling to near one while maintaining comparable short-horizon task performance.

Reproducibility Statement

The supplementary archive contains the scripts, fixed seeds, raw CSV/JSON logs, figure-generation scripts, and command manifest needed to regenerate the reported diagnostics and finite-step lookup table. Because the certificates depend on implementation details, the archive records the Sinkhorn dtype, positivity floor $\varepsilon_{\text{floor}}$, clipping/temperature settings, early-stopping tolerance, data path, and preprocessing. The accompanying code is available at `https://github.com/MaKosmakov/finite-step-sinkhorn-certificates`. The supplementary archive also includes saved final mixer stacks for the matched transformer run and the script `scripts/dobrushin_diagnostic.py`, which regenerates the depth-uniform Dobrushin audit and certificate-comparison tables. The script `scripts/budget_sensitivity_diagnostic.py` regenerates the post-hoc global-budget sensitivity table from the same saved stacks.

Ethics and Broader Impact Statement

This work studies finite-step normalization for nonnegative mixer products. The diagnostics use CIFAR-10 and do not involve human-subject data, personal data, or deployment decisions.

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A Certificate evaluation and additional diagnostics

For each layer, the implementation computes $\bar{\varepsilon}_\ell = \exp((\log B_1 - \sum_{j<\ell} \log(1+\varepsilon_c^{(j)}))/(L-\ell+1)) - 1$ and runs column-then-row Sinkhorn steps until either $\varepsilon_c^{(\ell)}(t) = \|\mathbf{1}^\top S^{(2t)} - \mathbf{1}^\top\|_{\infty,\mathrm{vec}} \le \bar{\varepsilon}_\ell$ or $t = t_{\max}$. The reported product certificate is $\exp(\sum_{\ell=1}^{L} \log(1+\varepsilon_c^{(\ell)}))$, and the exact realized backward gain is computed by the recursion $w_L = \mathbf{1}$, $w_{\ell-1} = (H_{\mathrm{res}}^{(\ell)})^\top w_\ell$, and $\|T_L\|_1 = \|w_0\|_{\infty,\mathrm{vec}}$. If any layer reaches $t_{\max}$ before satisfying its assigned tolerance, the adaptive budget certificate is marked as failed.

This appendix collects post-hoc certificate diagnostics and the stress-run audit plots.

A.1 Finite-horizon and depth-uniform Dobrushin diagnostic

To audit Proposition 3.4 on trained mixers, we saved the final per-layer residual mixer stacks from the matched transformer run and computed the finite-horizon Dobrushin certificate

$$1 + \frac{1}{2} \sum_{\ell=1}^{L} \left(\prod_{j=1}^{\ell-1} \delta_j \right) \|c_\ell\|_{1,\mathrm{vec}},$$

where $c_\ell = (H^{(\ell)})^\top \mathbf{1} - \mathbf{1}$ and $\delta_\ell = \delta_{\mathrm{D}}(H^{(\ell)})$. The corresponding depth-uniform bound follows by replacing the layerwise quantities by their maxima as in (3.11). The diagnostic uses the same final matched-run setting as Figure 1: three seeds, 5000 training images, 2000 test images, eight epochs, depth $L = 24$, and $n = 8$ streams.

Policy	L	n	$\bar{\delta}$	$\max_\ell \|c_\ell\|_1$	Product cert.	Dobrushin cert.	Exact $\|T_L\|_1$
softmax seed 0	24	8	9.267×10^{-4}	13.96	4.457×10^{21}	7.987	7.976
sinkhorn seed 0	24	8	0.2345	0.01692	1.0086	1.0001	1.0000
softmax seed 1	24	8	7.792×10^{-4}	13.96	4.460×10^{21}	7.987	7.985
sinkhorn seed 1	24	8	0.2082	0.01364	1.0053	1.0001	1.0000
softmax seed 2	24	8	9.942×10^{-4}	13.96	4.460×10^{21}	7.987	7.983
sinkhorn seed 2	24	8	0.2006	0.01372	1.0050	1.0001	1.0000

Table A.1: Post-hoc finite-horizon Dobrushin diagnostics for saved mixer stacks.

For the Sinkhorn-controlled mixers, the refined Dobrushin certificate is near one and close to the exact propagated gain. For the row-only softmax mixers, it is slightly smaller than the structural ceiling and again close to the exact gain. The product certificate remains the controller's directly allocated budget; the post-hoc Dobrushin calculation is the smallest reported audit bound for these saved stacks.

A.2 Additional matched-run diagnostics

The remaining matched-run diagnostics are shown in Figure A.1. The column-error panel records the finite-step marginal defect, and the gradient-norm panel is included as an auxiliary training diagnostic.

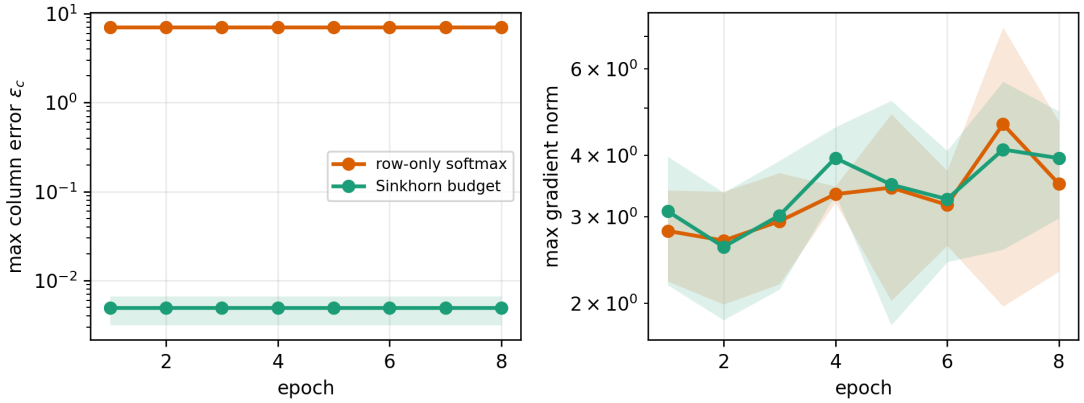


Figure A.1: Additional matched-run diagnostics. The column-error panel records the finite-step marginal defect, and the gradient-norm panel is included as an implementation audit.

A.3 Stress-run audit

These diagnostics use the same stress-run logs as the main certificate tables.

Dynamic-range audit. Table A.2 reports the final-epoch dynamic-range diagnostics for the stress run. For Sinkhorn, the fixed-cap column evaluates the same trained weights using $t = t_{\max}$. For the row-softmax baseline, κ_{obs} is a native pre-softmax diagnostic, not the clipped positive-matrix dynamic range used in the Sinkhorn certificate.

Table A.2: Final-epoch dynamic-range diagnostics for the stress run.

Model	med. κ_{obs}	std	med. $\kappa_{\text{obs}}^{\text{fix}}$	std
softmax	8556629.04	840682.39	—	—
sinkhorn (early)	14489.91	1315.73	14487.00	1324.17

Table A.3: Final-epoch stack certificate diagnostics. The exact backward gain is computed by vector propagation, while the certificate is $\prod_\ell (1 + \varepsilon_c^{(\ell)})$.

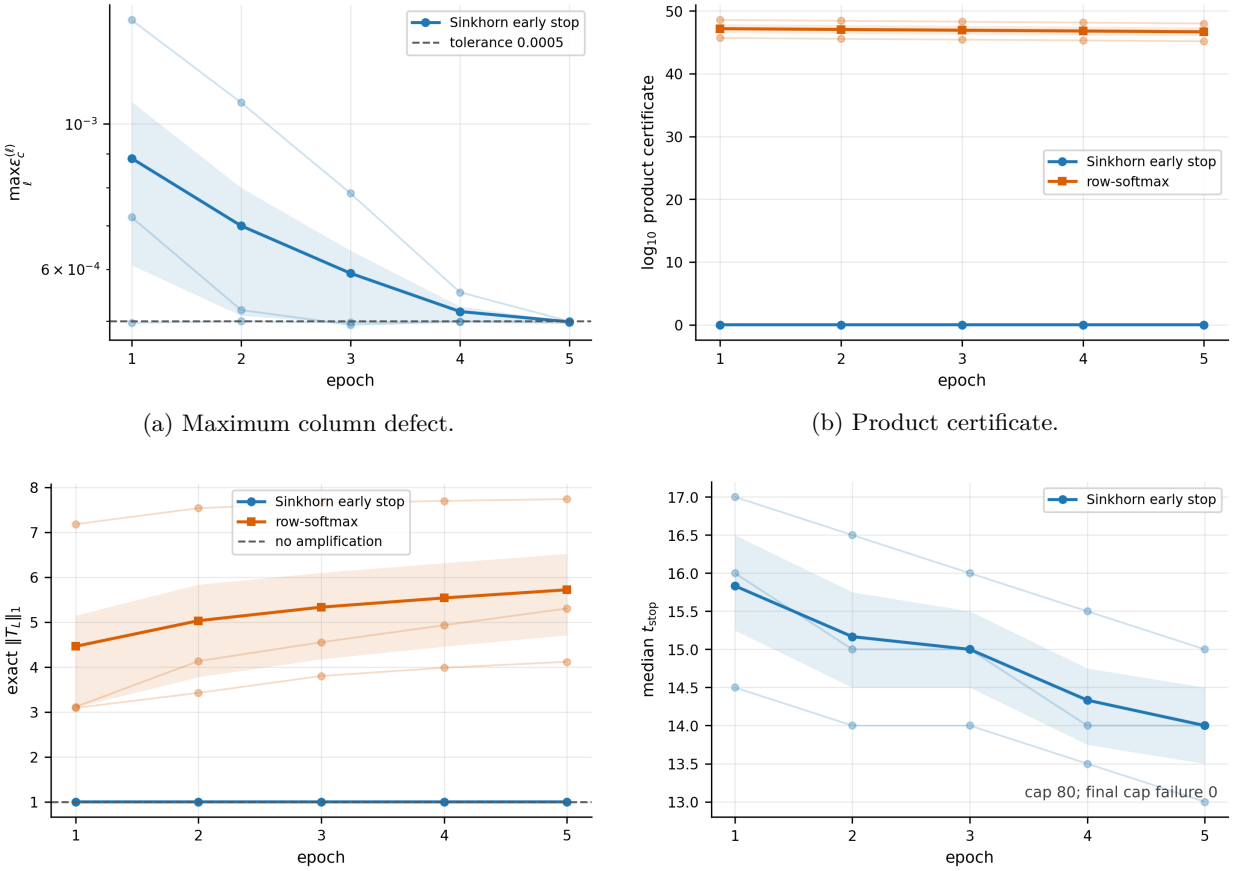
Model	cert.	exact $\|T_L\|_1$	exact/cert.
softmax	$3.61{\times}10^{47} \pm 5{\times}10^{47}$	5.72 ± 2	—
sinkhorn (early)	1.05 ± 0.0009	$1 \pm 8{\times}10^{-5}$	0.951 ± 0.0009

For the early-stopped Sinkhorn run, the final epoch reaches the tolerance $\varepsilon_{\text{tol}} = 5 \times 10^{-4}$ with median stopping time $t_{\text{stop}} = 14$. Re-evaluating the same trained weights with the full cap $t_{\max} = 80$ gives a smaller column defect, about 1.9×10^{-4}, confirming that the reported stopping behavior is tolerance-driven rather than cap-limited.

The row-softmax baseline illustrates that row stochasticity alone does not give a useful backward product certificate: the product certificate is enormous, although the realized gain is structurally capped and remains $O(n)$. The Sinkhorn controller keeps both the certificate and the exact propagated gain near one.

For the Sinkhorn run, the associated local diagnostic is $\rho_* = 0.631 \pm 0.01$, and the cap-failure rate is zero.

Stress-run audit trajectories. Figure A.2 summarizes the main stress-run diagnostics over training. The maximum column defect stays near the early-stopping tolerance, the product certificate remains below the prescribed budget, and the exact propagated gain stays close to one. The stopping-count panel records the compute savings from early stopping. Cap failures are zero in the reported run.



(a) Maximum column defect.

(b) Product certificate.

(c) Exact propagated gain.

(d) Early-stop step counts.

Figure A.2: Stress-run audit diagnostics for the finite-step Sinkhorn controller. These plots use the same logs as the main certificate tables.

B Auxiliary finite-step estimates

Proof of Lemma 4.1. Let $a := \max_i u_i$, $b := \min_i u_i$, and $r := a/b = e^{\delta} \geq 1$. The mean constraint gives $b \leq 1 \leq a$. For the upper deviation, fixing the ratio $a/b = r$ and the mean, the largest possible value of a occurs when one entry is a and the remaining $n-1$ entries are b. Thus

$$a \leq \frac{nr}{r+n-1}, \qquad a - 1 \leq \frac{(n-1)(r-1)}{r+n-1}.$$

For the lower deviation, the smallest possible value of b occurs when one entry is b and the remaining $n-1$ entries are a. Thus

$$b \geq \frac{n}{1+(n-1)r}, \qquad 1 - b \leq \frac{(n-1)(r-1)}{1+(n-1)r}.$$

Since $1 + (n-1)r \geq r + n - 1$ for $n \geq 2$ and $r \geq 1$, the lower-tail bound is no larger than the upper-tail bound. Therefore

$$\|u - \mathbf{1}\|_{\infty,\mathrm{vec}} \leq \frac{(n-1)(r-1)}{r+n-1} = \Psi_n(\delta).$$

Solving $\Psi_n(\delta) \le \varepsilon$ for $r = e^\delta$ gives (4.15).

Proof of Lemma 4.2. Since $z = \Pi \log u$ and $n^{-1}\sum_i u_i = 1$, there is a unique scalar normalization such that

$$u_i = \frac{ne^{z_i}}{\sum_{k=1}^n e^{z_k}}.$$

We first maximize a single coordinate. Under the constraints $\sum_i z_i = 0$ and $\|z\|_2 \le \eta$, the largest possible value of one coordinate is

$$a = \sqrt{\frac{n-1}{n}}\,\eta,$$

attained when the remaining coordinates all equal $-a/(n-1)$. Indeed, if $z_i = a$ is fixed, the sum of the other coordinates is $-a$; Jensen's inequality gives

$$\sum_{k\neq i} e^{z_k} \ge (n-1)e^{-a/(n-1)},$$

with equality exactly in the equal-tail configuration. Therefore

$$u_i \le \frac{ne^a}{e^a + (n-1)e^{-a/(n-1)}}.$$

This gives the upper-tail deviation $\Gamma_n^+(\eta)$.

For the lower tail, two complementary estimates are useful. First, let $M = \max_i z_i$ and $m = \min_i z_i$. For every coordinate,

$$u_i = \frac{ne^{z_i}}{\sum_k e^{z_k}} \ge \frac{ne^m}{ne^M} = e^{-(M-m)}.$$

Since

$$M - m = \langle z, e_{i_{\max}} - e_{i_{\min}}\rangle \le \sqrt{2}\,\|z\|_2 = \sqrt{2}\,\eta,$$

we recover the global estimate

$$1 - u_i \le 1 - e^{-\sqrt{2}\eta}.$$

To obtain the correct first-order behavior, Cauchy–Schwarz and $\sum_i z_i = 0$ give $|z_i| \le a = \sqrt{(n-1)/n}\,\eta$ for every coordinate. The elementary Taylor remainder bound

$$e^x \le 1 + x + \frac{1}{2}e^{|x|}x^2$$

therefore gives

$$\frac{1}{n}\sum_{k=1}^n e^{z_k} \le 1 + \frac{e^a}{2n}\sum_{k=1}^n z_k^2 = 1 + \frac{e^a}{2n}\eta^2.$$

Together with $z_i \ge -a$, this gives

$$u_i = \frac{e^{z_i}}{n^{-1}\sum_k e^{z_k}} \ge \frac{e^{-a}}{1 + \frac{e^a}{2n}\eta^2}.$$

Taking the better of the two valid lower bounds gives the stated $\Gamma_n^-(\eta)$. Since both $\Gamma_n^+(\eta)$ and $\Gamma_n^-(\eta)$ have first-order coefficient $\sqrt{(n-1)/n}$, equation (4.18) follows.

B.1 Local finite-orbit derivative envelope

With the notation of Section 4.4, consider the derivative on a neighborhood of the Sinkhorn limit.

For $x \in \mathbf{1}^\perp$, define

$$g(x) := P^\top e^x, \qquad a(x) := P(g(x)^{-1}),$$

and

$$J_F(x) := \Pi \operatorname{diag}(a(x))^{-1} P \operatorname{diag}(g(x))^{-2} P^\top \operatorname{diag}(e^x) \Pi, \tag{B.1}$$

$$J_L(x) := \Pi \operatorname{diag}(g(x))^{-1} P^\top \operatorname{diag}(e^x) \Pi. \tag{B.2}$$

For $R > 0$, set

$$\rho_R := \sup_{\substack{x \in \mathbf{1}^\perp \\ \|x\|_2 \le R}} \|J_F(x)\|_{2\to 2}, \qquad \lambda_R := \sup_{\substack{x \in \mathbf{1}^\perp \\ \|x\|_2 \le R}} \|J_L(x)\|_{2\to 2}. \tag{B.3}$$

Lemma B.1 (Local spectral envelope)**.** *Assume $\rho_R < 1$ and $\|x_0\|_2 \le R$. Then the Sinkhorn orbit remains in $\|x\|_2 \le R$ and, for every $t \ge 1$,*

$$\|z_t\|_2 \le \lambda_R \rho_R^{t-1} \|x_1 - x_0\|_2. \tag{B.4}$$

Therefore,

$$\varepsilon_c^{(2t)} \le \Gamma_n\left(\lambda_R \rho_R^{t-1} \|x_1 - x_0\|_2\right). \tag{B.5}$$

The same bounds hold with certified upper bounds $\widehat{\rho}_R \ge \rho_R$, $\widehat{\lambda}_R \ge \lambda_R$, provided $\widehat{\rho}_R < 1$.

Proof. The displayed formulae for J_F and J_L are obtained by differentiating (4.29). Because $\mathcal{F}(0) = 0$, the mean-value theorem gives, for any $x \in \mathbf{1}^\perp$ with $\|x\|_2 \le R$,

$$\|\mathcal{F}(x)\|_2 \le \rho_R \|x\|_2.$$

Thus the ball is invariant whenever $\rho_R < 1$. Applying the same mean-value estimate to the differences gives

$$\|x_{s+1} - x_s\|_2 \le \rho_R \|x_s - x_{s-1}\|_2, \qquad s \ge 1,$$

and therefore

$$\|x_t - x_{t-1}\|_2 \le \rho_R^{t-1} \|x_1 - x_0\|_2.$$

Since $z_t = \mathcal{L}(x_t) - \mathcal{L}(x_{t-1})$, another mean-value estimate gives (B.4). Lemma 4.2 converts the centered logarithmic error into the ordinary column-sum error. □

Orbit-wise derivative bound. Let $x_0, x_1, \ldots, x_t$ be the actual logarithmic Sinkhorn orbit and define

$$\rho_s := \sup_{0 \le \theta \le 1} \left\| J_F\big((1-\theta)x_{s-1} + \theta x_s\big) \right\|_{2\to 2}, \qquad s \ge 1,$$

$$\lambda_t^{\mathrm{seg}} := \sup_{0 \le \theta \le 1} \left\| J_L\big((1-\theta)x_{t-1} + \theta x_t\big) \right\|_{2\to 2}.$$

Then

$$\|z_t\|_2 \le \lambda_t^{\mathrm{seg}} \left(\prod_{s=1}^{t-1} \rho_s \right) \|x_1 - x_0\|_2, \tag{B.6}$$

and thus

$$\varepsilon_c^{(2t)} \le \Gamma_n\left(\lambda_t^{\mathrm{seg}} \left(\prod_{s=1}^{t-1} \rho_s \right) \|x_1 - x_0\|_2 \right). \tag{B.7}$$

This a posteriori bound replaces the global dynamic-range rate by derivative norms along the realized orbit. Applying the mean-value theorem on each orbit segment and then Lemma 4.2 gives the stated bound.

C Predictor expansion and full-block bounds

C.1 Second-order local predictor formulas

The second-order diagnostic predictor keeps the quadratic terms in the logarithmic Sinkhorn map. With $K = PP^\top$, define

$$Q_F(x) := -P\big((P^\top x)^{\odot 2}\big) + \frac{1}{2}K(x^{\odot 2}) + \frac{1}{2}(Kx)^{\odot 2}, \tag{C.1}$$

$$Q_L(x) := \frac{1}{2}P^\top(x^{\odot 2}) - \frac{1}{2}(P^\top x)^{\odot 2}. \tag{C.2}$$

Then

$$\mathcal{F}(x) = Kx + \Pi Q_F(x) + O(\|x\|^3), \qquad \mathcal{L}(x) = \Pi P^\top x + \Pi Q_L(x) + O(\|x\|^3). \tag{C.3}$$

Set

$$x_t^{[1]} := K^t x_0, \qquad x_t^{[2]} := K^t x_0 + \sum_{s=0}^{t-1} K^{t-1-s}\Pi Q_F(K^s x_0). \tag{C.4}$$

The second-order centered log-column predictor is

$$\ell_t^{[2]} := \Pi\Big[P^\top x_t^{[2]} + Q_L(x_t^{[1]}) - P^\top x_{t-1}^{[2]} - Q_L(x_{t-1}^{[1]})\Big]. \tag{C.5}$$

We then define

$$\widehat{v}_t^{[2]} := \frac{n\exp(\ell_t^{[2]})}{\sum_{j=1}^n \exp((\ell_t^{[2]})_j)}, \qquad \widehat{\varepsilon}_{c,2}^{(2t)} := \|\widehat{v}_t^{[2]} - \mathbf{1}\|_{\infty,\mathrm{vec}}. \tag{C.6}$$

For each fixed t,

$$z_t = \ell_t^{[2]} + O_t(\|x_0\|^3). \tag{C.7}$$

Thus (C.6) is the finite-step prediction to compare with early numerical points, while (4.40) gives the asymptotic slope.

C.2 Full-layer Jacobian $A_{\max}$ bounds

The mixer-product certificates in Section 3 isolate the contribution of stream mixing. Full-network certificates additionally require independent control of the nonlinear residual branch. We first record the generic composition bound, then give the row-major Jacobian calculation for the mHC-style layer in (2.1).

Proposition C.1 (Full-block $A_{\max}$ composition bound)**.** *Consider a depth-L network of the form*

$$X_\ell = H^{(\ell)}X_{\ell-1} + \mathcal{F}_\ell(X_{\ell-1}), \qquad \ell = 1,\dots,L, \tag{C.8}$$

where $H^{(\ell)} \in \mathbb{R}^{n\times n}$ mixes streams and $\mathcal{F}_\ell$ denotes the remaining nonlinear part of the block. Suppose

$$A_{\max}(\mathcal{F}_\ell(X) - \mathcal{F}_\ell(Y)) \le a_\ell A_{\max}(X - Y) \tag{C.9}$$

for all relevant X, Y. Then

$$A_{\max}(X_L - Y_L) \le \prod_{\ell=1}^{L}\big(\|H^{(\ell)}\|_\infty + a_\ell\big)A_{\max}(X_0 - Y_0). \tag{C.10}$$

If, in addition, along the realized trajectory the adjoint Jacobian of the nonlinear branch satisfies

$$A_{\max}(D\mathcal{F}_\ell(X_{\ell-1})^* G) \le b_\ell A_{\max}(G), \tag{C.11}$$

then the input-gradient amplification obeys

$$A_{\max}(\nabla_{X_0}\mathcal{L}) \le \prod_{\ell=1}^{L}\big(\|H^{(\ell)}\|_1 + b_\ell\big)A_{\max}(\nabla_{X_L}\mathcal{L}). \tag{C.12}$$

In particular, if $H^{(\ell)} \geq 0$ *is* $(\varepsilon_r^{(\ell)}, \varepsilon_c^{(\ell)})$*-doubly stochastic, then*

$$A_{\max}(\nabla_{X_0}\mathcal{L}) \leq \prod_{\ell=1}^{L}\big(1+\varepsilon_c^{(\ell)}+b_\ell\big)A_{\max}(\nabla_{X_L}\mathcal{L}). \tag{C.13}$$

Proof. Subtracting two trajectories and using the triangle inequality gives

$$\begin{aligned} A_{\max}(X_\ell - Y_\ell) &\leq A_{\max}\big(H^{(\ell)}(X_{\ell-1}-Y_{\ell-1})\big) + A_{\max}\big(\mathcal{F}_\ell(X_{\ell-1})-\mathcal{F}_\ell(Y_{\ell-1})\big) \\ &\leq \big(\|H^{(\ell)}\|_\infty + a_\ell\big)A_{\max}(X_{\ell-1}-Y_{\ell-1}), \end{aligned}$$

which iterates to (C.10). The backward recursion has the form

$$G_{\ell-1} = (H^{(\ell)})^\top G_\ell + D\mathcal{F}_\ell(X_{\ell-1})^* G_\ell.$$

Using (2.5) and (C.11) gives

$$A_{\max}(G_{\ell-1}) \leq \big(\|H^{(\ell)}\|_1 + b_\ell\big)A_{\max}(G_\ell),$$

and iteration gives (C.12). The marginal version follows from (3.1). □

Let $X \in \mathbb{R}^{n\times C}$ and define the row-major vectorization

$$\mathrm{vec}_r(X) := \mathrm{vec}(X^\top) \in \mathbb{R}^{nC}, \tag{C.14}$$

where vec is the standard column-stacking operator. With this convention one has, whenever products are well-defined,

$$\mathrm{vec}_r(AXB) = (A \otimes B^\top)\,\mathrm{vec}_r(X), \tag{C.15}$$

and in particular

$$\mathrm{vec}_r(HX) = (H \otimes I_C)\,\mathrm{vec}_r(X). \tag{C.16}$$

Layer Jacobian and induced-norm bounds Fix a layer ℓ. Let $H_{\mathrm{res}}^{(\ell)} \in \mathbb{R}^{n\times n}$ be the residual mixer and $H_{\mathrm{pre}}^{(\ell)}, H_{\mathrm{post}}^{(\ell)} \in \mathbb{R}^{1\times n}$ the pre/post aggregation row vectors. Let $F^{(\ell)} : \mathbb{R}^C \to \mathbb{R}^C$ denote the feature map applied to the aggregated feature vector $u := H_{\mathrm{pre}}^{(\ell)} X \in \mathbb{R}^C$, and let $J_F^{(\ell)} \in \mathbb{R}^{C\times C}$ be the Jacobian of $F^{(\ell)}$ with respect to u.

The layer update is

$$X \mapsto H_{\mathrm{res}}^{(\ell)} X + H_{\mathrm{post}}^{(\ell)\top} F^{(\ell)}(H_{\mathrm{pre}}^{(\ell)} X). \tag{C.17}$$

Applying Equation (C.15) and the chain rule gives the Jacobian (with respect to $\mathrm{vec}_r(X)$):

$$\begin{aligned} J^{(\ell)} &= H_{\mathrm{res}}^{(\ell)} \otimes I_C + \big(H_{\mathrm{post}}^{(\ell)\top} \otimes I_C\big)\, J_F^{(\ell)}\, \big(H_{\mathrm{pre}}^{(\ell)} \otimes I_C\big) \\ &= H_{\mathrm{res}}^{(\ell)} \otimes I_C + \big(H_{\mathrm{post}}^{(\ell)\top} H_{\mathrm{pre}}^{(\ell)}\big) \otimes J_F^{(\ell)}, \end{aligned} \tag{C.18}$$

where the second line uses $J_F^{(\ell)} = [1] \otimes J_F^{(\ell)}$ and the mixed-product identity for Kronecker products.

Proposition C.2 (Layer Jacobian induced-norm bounds)**.** *With the notation above,*

$$\left\|J^{(\ell)}\right\|_\infty \leq \left\|H_{\mathrm{res}}^{(\ell)}\right\|_\infty + \left\|H_{\mathrm{post}}^{(\ell)}\right\|_{\infty,\mathrm{vec}} \left\|J_F^{(\ell)}\right\|_\infty \left\|H_{\mathrm{pre}}^{(\ell)}\right\|_{1,\mathrm{vec}}. \tag{C.19}$$

Similarly,

$$\left\|J^{(\ell)}\right\|_1 \leq \left\|H_{\mathrm{res}}^{(\ell)}\right\|_1 + \left\|H_{\mathrm{post}}^{(\ell)}\right\|_{1,\mathrm{vec}} \left\|J_F^{(\ell)}\right\|_1 \left\|H_{\mathrm{pre}}^{(\ell)}\right\|_{\infty,\mathrm{vec}}. \tag{C.20}$$

Proof. The Kronecker factorization in (C.18) gives an exact norm identity for the nonlinear rank-one stream term. Namely,

$$\begin{aligned}\left\|\left(H_{\text{post}}^{(\ell)\top} H_{\text{pre}}^{(\ell)}\right) \otimes J_F^{(\ell)}\right\|_\infty &= \|H_{\text{post}}^{(\ell)\top} H_{\text{pre}}^{(\ell)}\|_\infty \|J_F^{(\ell)}\|_\infty \\ &= \|H_{\text{post}}^{(\ell)}\|_{\infty,\text{vec}} \|H_{\text{pre}}^{(\ell)}\|_{1,\text{vec}} \|J_F^{(\ell)}\|_\infty.\end{aligned}$$

Together with $\|H_{\text{res}}^{(\ell)} \otimes I_C\|_\infty = \|H_{\text{res}}^{(\ell)}\|_\infty$ and the triangle inequality, this proves (C.19). Similarly,

$$\left\|\left(H_{\text{post}}^{(\ell)\top} H_{\text{pre}}^{(\ell)}\right) \otimes J_F^{(\ell)}\right\|_1 = \|H_{\text{post}}^{(\ell)}\|_{1,\text{vec}} \|H_{\text{pre}}^{(\ell)}\|_{\infty,\text{vec}} \|J_F^{(\ell)}\|_1,$$

which gives (C.20). The only remaining slack in these layerwise bounds is the triangle inequality between the residual and nonlinear Jacobian terms. □

Corollary C.3 (Conditional full-block Jacobian certification)**.** *Assume each residual mixer is entrywise nonnegative and* $(\varepsilon_r^{(\ell)}, \varepsilon_c^{(\ell)})$*-doubly stochastic, so that* $\left\|H_{\text{res}}^{(\ell)}\right\|_\infty \le 1 + \varepsilon_r^{(\ell)}$ *and* $\left\|H_{\text{res}}^{(\ell)}\right\|_1 \le 1 + \varepsilon_c^{(\ell)}$*. If, in addition, there exist constants* $b_{\text{pre}}, b_{\text{post}}, L_F \ge 0$ *such that for all layers* ℓ*,*

$$\left\|H_{\text{pre}}^{(\ell)}\right\|_{1,\text{vec}} \le b_{\text{pre}}, \qquad \left\|H_{\text{post}}^{(\ell)}\right\|_{\infty,\text{vec}} \le b_{\text{post}}, \qquad \left\|J_F^{(\ell)}\right\|_\infty \le L_F, \tag{C.21}$$

then

$$\left\|J^{(\ell)}\right\|_\infty \le (1 + \varepsilon_r^{(\ell)}) + b_{\text{post}}\, L_F\, b_{\text{pre}}. \tag{C.22}$$

Thus

$$\left\|J^{(L)} \cdots J^{(1)}\right\|_\infty \le \prod_{\ell=1}^{L} \left((1 + \varepsilon_r^{(\ell)}) + b_{\text{post}}\, L_F\, b_{\text{pre}}\right). \tag{C.23}$$

The backward induced-norm product has the parallel bound

$$\left\|J^{(L)} \cdots J^{(1)}\right\|_1 \le \prod_{\ell=1}^{L} \left(1 + \varepsilon_c^{(\ell)} + \left\|H_{\text{post}}^{(\ell)}\right\|_{1,\text{vec}} \left\|J_F^{(\ell)}\right\|_1 \left\|H_{\text{pre}}^{(\ell)}\right\|_{\infty,\text{vec}}\right). \tag{C.24}$$

Here L_F is an architecture-dependent bound on the nonlinear block Jacobian. For a pointwise MLP, it can be estimated from the induced norms of the affine maps and the Lipschitz constants of the activations. For an attention block, it also depends on the projection operators and the data-dependent attention map. The full-block certificate therefore retains L_F explicitly rather than assigning it a universal value.